Multi-Class Annotation Errors in Causal Inference: Topology-Dependent Bias, Sensitivity, and Correction

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Abstract

Large language models (LLMs) are increasingly used as annotators in computational social science, yet annotation errors propagate into downstream causal estimates in ways that binary measurement error theory cannot predict. We present a systematic framework characterizing how K-class confusion matrix structure interacts with DAG topology to determine bias direction and magnitude. We derive bias expressions for seven canonical DAG topologies under multi-class non-differential misclassification and establish that five topologies show no sign reversal across extensive numerical verification (proven analytically for confounding and collider via sufficient conditions; Appendix A), while exposure and instrumental variable topologies can produce sign reversal—under realistic classifier accuracy (diagonal ≥ 0.7), the violation rate is 13.7% with 2.1% sign reversal; across the full Dirichlet$(1, 1, 1)$ parameter space, the rate reaches 83.4%. We introduce Annotation Sensitivity Values (ASVs) with three complementary measures revealing a topology-dependent vulnerability spectrum from IV (maximally fragile, 52.6% sign reversal) to M-bias (immune). We evaluate five correction methods and prove a Frisch–Waugh–Lovell invariance result: matrix-based corrections achieve 0% bias reduction for four topologies, while PPI++ achieves 47.0% RMSE reduction but requires gold labels and is harmful for M-bias (−88%). Empirical validation on three tasks (K=2, 3, 10) with four LLMs confirms structural predictions: zero sign reversals across five guaranteed topologies, while 61% of exposure configurations exhibit coefficient-level sign reversal in secondary coefficients. Topology-aware analysis is necessary for valid causal inference with LLM-generated annotations.

1 Introduction

Large language models are rapidly replacing human annotators in computational social science, with researchers using LLM-generated labels for tasks ranging from sentiment classification to political stance detection (Egami et al., 2023; Baumann et al., 2025). Crucially, most of these tasks are inherently multi-class: stance detection uses 3–4 categories, sentiment analysis 5, and disaster response taxonomies 10 or more. This shift promises dramatic cost reduction, but introduces a critical validity gap: annotation errors from imperfect classifiers propagate into downstream causal estimates in ways that depend on both the error structure and the causal design. Baumann et al. (2025) demonstrate that across 37 annotation tasks, 31% of hypotheses yield incorrect conclusions with state-of-the-art LLMs, rising to 50% for smaller models; these errors are highly correlated across models (60% agreement when both err) (Kim et al., 2025), undermining ensemble-based mitigation. Analyzing the *structure* of annotation error, rather than simply reducing its magnitude, is essential for valid downstream inference.

The consequences can be catastrophic. Even under realistic classifier accuracy (diagonal ≥ 0.7), 13.7% of K=3 confusion matrices produce violations of the classical binary guarantee, with 2.1% causing outright *sign reversal* of the estimated treatment effect. Across the full Dirichlet$(1, 1, 1)$ parameter space, sign reversal reaches 83.4%, establishing that it is the majority outcome under uniform priors, not a pathological edge case. This failure breaks the classical binary guarantee that non-differential misclassification biases estimates "toward the null" (Brenner, 1993), a guarantee that many applied researchers implic-

itly rely upon. Moreover, standard classification metrics do not predict downstream causal risk: within a narrow F_1 range of 0.614–0.690 across three frontier LLMs, relative bias varies by an order of magnitude across topologies (from 4.1% to 42.1%), illustrating that *topology*, not annotator quality, determines downstream risk.

The mechanism is algebraic, and the transition at $K \geq 3$ is a qualitative break, not a gradual degradation. For $K=2$, the stochastic constraint forces any confusion matrix with $C_{kk} > 0.5$ to preserve coefficient signs—an algebraic property of the 2×2 matrix dimension, not of a specific error rate. This guarantee *vanishes* at $K \geq 3$: the higher-dimensional confusion matrix admits negative effective weights on categories with opposite-sign coefficients, enabling sign reversal even when every diagonal entry exceeds 0.5 (Veierød and Laake, 2001). This fundamental asymmetry has received limited systematic treatment. Brenner (1993) identified the possibility of polytomous reversal but did not characterize it across causal designs. Lee and Wood-Doughty (2024) extended classifier error analysis to categorical variables and LLM classifiers, but covered only one DAG topology (unobserved confounding). No existing framework addresses how the same confusion matrix produces qualitatively different bias across different causal identification strategies.

We address these gaps with three contributions:

1. **Multi-class bias taxonomy.** We derive bias expressions for seven canonical DAG topologies (Figure 1) under K-class nondifferential misclassification and establish that five topologies maintain a structural sign-flip guarantee while exposure and IV can produce sign reversal. Binarization amplifies the gap: collapsing $K=3$ to binary introduces additional bias varying 154× across topologies.

2. **Annotation Sensitivity Values (ASVs).** We propose three complementary sensitivity measures—magnitude, significance, and sign-flip ASV—revealing a vulnerability spectrum from IV (maximally fragile, 52.6% sign reversal) to M-bias (immune, ASV $= \infty$).

3. **Topology-aware correction guidance.** We evaluate five correction methods and show that matrix-based corrections achieve exactly 0% bias reduction for four topologies due to Frisch–Waugh–Lovell invariance. PPI++ achieves 47.0% RMSE reduction but requires gold labels and is harmful for M-bias (−88%).

2 Related Work

Measurement error in causal inference. The statistical foundations are established in Carroll et al. (2006) and Gustafson (2004), with starred-node DAGs formalized by Hernán and Cole (2009). For binary variables, nondifferential misclassification biases toward the null (Brenner, 1993; Ogburn and VanderWeele, 2012), but Dosemeci et al. (1990) and Jurek et al. (2005) showed this fails for polytomous exposures. Veierød and Laake (2001) derived multi-category bias expressions for Poisson regression, and Edwards et al. (2021) showed dependent misclassification can bias in either direction. All of these works analyze a single DAG topology; our taxonomy is the first to characterize how the *same* confusion matrix produces qualitatively different bias across seven topologies.

Text classifiers and LLM annotation. Grimmer et al. (2022) provide the foundation for text-as-data; Feder et al. (2022) offer a comprehensive taxonomy of causal inference in NLP, categorizing text-as-treatment and text-as-confounder paradigms. Wood-Doughty et al. (2018) first applied DAG reasoning to classifier error. TeBlunthuis et al. (2024) gave the most complete binary treatment (four DAG scenarios, $K=2$), and Lee and Wood-Doughty (2024) extended to categorical LLM classifiers but covered only confounding. On annotation quality, Baumann et al. (2025) found 31% incorrect conclusions with LLM labels, Cheng et al. (2024) propose structured approaches to reduce LLM measurement error, and Kim et al. (2025) showed 60% error correlation across models. Audinet de Pieuchon et al. (2025) benchmark debiasing methods for LLM-generated parameter estimates; Keith et al. (2020) survey the broader text-and-causal-inference landscape. These works

either restrict to binary labels, cover a single topology, or diagnose quality without connecting it to causal validity. Our framework provides the missing link.

Sensitivity analysis and correction methods. Our ASV framework draws on E-values for unmeasured confounding (VanderWeele and Ding, 2017) and probabilistic bias analysis (Lash et al., 2009). Unlike E-values, ASVs comprise three types (magnitude, significance, sign-flip) because failure modes are topology-dependent; unlike PBA, ASVs provide monotonicity guarantees over $K \times K$ matrices. For correction, PPI++ (Angelopoulos et al., 2023) and DSL (Egami et al., 2023) are designed as topology-agnostic; Zrnić and Candès (2024) extend PPI to gold-free cross-prediction settings, and Gligorić et al. (2025) leverage LLM confidence scores for inference. Our contribution reveals that this agnosticism is costly: correction effectiveness is itself topology-dependent, with matrix-based methods provably ineffective for four topologies.

3 Method

3.1 Setup and Notation

Let A denote a true K-class categorical variable (e.g., sentiment with $K{=}3$ levels) and A^* its observed, potentially misclassified version produced by an LLM annotator. The misclassification process is characterized by a $K \times K$ confusion matrix C with entries $C_{jk} = P(A^*{=}j \mid A{=}k)$, where each column sums to one. Following Hernán and Cole (2009), we represent this measurement process using *starred-node* DAGs: the true variable A is latent, and the observed A^* is connected to A via a dashed measurement edge.

Definition 1 (Non-differential misclassification)**.** *Misclassification of A is non-differential with respect to outcome Y and covariates $\mathbf{Z}$ if $P(A^*{=}j \mid A{=}k, Y, \mathbf{Z}) = P(A^*{=}j \mid A{=}k) = C_{jk}$ for all j, k.*

We study seven canonical DAG topologies (Figure 1) (Pearl, 2009) in which A occupies structurally distinct causal roles: confounder, exposure, mediator, collider, M-bias variable, front-door mediator, and instrument. Each role induces a different functional relationship between C and the probability limit of the standard causal estimator, yielding topology-dependent bias.

3.2 Multi-Class Bias Expressions

For each topology, we derive the probability limit of the naive estimator that substitutes A^* for A. Let τ denote the target causal parameter and $\hat{\tau}(A^*)$ the estimator using misclassified annotations.

Confounding (A = confounder). When A confounds the $X \to Y$ relationship, the OLS estimator controlling for A^* has probability limit

$$\operatorname{plim} \hat{\tau}(A^*) = \big(\mathbb{E}[Z'Z]\big)^{-1}\mathbb{E}[Z'X]\,\boldsymbol{\beta}, \qquad (1)$$

where $Z = [X, \text{dummies}(A^*)]$, $\boldsymbol{\beta}$ contains the true coefficients, and the target parameter τ is the coefficient on X. Misclassification produces residual confounding that monotonically approaches the omitted variable bias, but the sign of $\hat{\tau}$ is preserved: $\tau_{\text{sign_flip}} = 0$ structurally.[1]

Exposure (A = treatment variable). When A is the exposure of interest, the biased coefficient for category j becomes a weighted combination of true coefficients:

$$\operatorname{plim} \hat{\beta}_j(A^*) = \sum_{k=1}^{K} w_{jk}\,\beta_k, \qquad (2)$$

where w_{jk} depends on the confusion matrix C and the marginal distribution of A. Unlike the binary case, where non-differential misclassification guarantees attenuation toward the null (Brenner, 1993), the $K{\geq}3$ case permits w_{jk} to assign negative weight to categories with opposite-sign coefficients, producing *sign reversal*. A near-permutation confusion matrix effectively swaps category labels, making sign reversal the dominant outcome (83.4% under uniform Dirichlet prior). This

[1] This result is consistent with Brenner (1993), who showed that the adjusted estimate with misclassified polytomous confounders can be biased toward or away from the null. Both findings hold simultaneously: misclassification degrades confounding control, producing residual bias whose magnitude can exceed the correctly adjusted estimate (amplification), but the treatment effect sign is never reversed. The distinction is between *sign reversal* ($\text{sign}(\hat{\tau}) \neq \text{sign}(\tau)$, which we show never occurs) and *amplification* ($|\hat{\tau}| > |\tau|$, which occurs universally in this topology).

failure mode is structurally impossible at $K{=}2$ because a $2{\times}2$ stochastic matrix with positive diagonal cannot reverse the sign of a single coefficient.

Mediation (A = mediator). When A mediates the $X{\to}Y$ relationship, the direct effect estimator controlling for A^* has probability limit

$$\operatorname{plim} \hat{\tau}_{\text{direct}}(A^*) = \tau_{\text{direct}} + \big(1 - \lambda(C)\big)\, \tau_{\text{indirect}}, \tag{3}$$

where $\lambda(C) \in [0, 1]$ summarizes attenuation (Appendix A, Proposition 3), with $\lambda(I) = 1$ under perfect classification. Misclassification ($\lambda < 1$) causes the residual indirect effect to be absorbed into the direct path estimate, producing upward bias in $|\hat{\tau}_{\text{direct}}|$ but preserving the sign ($\tau_{\text{sign_flip}} = 0$).

Collider (A = collider, conditioned on A^*). Conditioning on a collider A of X and Y opens a spurious path. When we condition on A^* instead, misclassification attenuates this spurious association because A^* is a noisy proxy for A. The collider topology thus exhibits a *protective* property: annotation error reduces rather than amplifies the bias from collider stratification, with $\tau_{\text{sign_flip}} = 0$.

M-bias (A = intermediate collider). In the M-shaped DAG ($U_1 \to X$, $U_1 \to A$, $U_2 \to A$, $U_2 \to Y$, $X \to Y$), adjusting for A introduces a small spurious association via the M-bias path (Greenland, 2003). Misclassifying A weakens this adjustment, reducing the M-bias. This topology is *immune*: $\hat{\tau}(A^*)$ is closer to τ than $\hat{\tau}(A)$ for all non-trivial confusion matrices, yielding ASV $= \infty$.

Front-door (A = front-door mediator). In the front-door DAG ($X \to A \to Y$ with unmeasured $U \to X$, $U \to Y$), the front-door estimator (Bellemare et al., 2024) uses A to identify effects despite unmeasured confounding. Misclassification attenuates both $X \to A$ and $A \to Y$ relationships, producing moderate bias with $\tau_{\text{sign_flip}} = 0$.

Instrumental variable (A = instrument). The Wald estimator (Angrist and Imbens, 1995) $\hat{\tau}_{\text{IV}} = \hat{\text{cov}}(A^*, Y)/\hat{\text{cov}}(A^*, X)$ has C in the denominator. Misclassification can drive first-stage coefficients toward zero or reverse their sign, causing the ratio estimator to diverge or flip. Under random Dirichlet matrices, 52.6% produce sign reversal, making IV the most fragile topology.

Structural sign-flip guarantee.

Proposition 1 (Sign preservation—analytic results)**.** *Under non-differential misclassification with $C_{kk} > 0$ for all k:* (a) *Confounding preserves sign when* $\text{sign}(\tau) = \text{sign}(\tau{+}B_{\text{OV}})$, *where* $B_{\text{OV}}{=}\,\text{Cov}(g(A), f(A))/\,\text{Var}(T)$ *is the omitted-variable bias (Cauchy–Schwarz; Appendix A.3).* (b) *Collider preserves sign when* $|\tau| > \sqrt{\text{Var}(g)\,\text{Var}(h)}/\mathbb{E}[\text{Var}(T|A)]$ *(Berkson attenuation; Appendix A.6).* (c) *Exposure can produce sign reversal for $K{\geq}3$, even with $C_{kk}{>}0.5$ (counterexample in Appendix A.8).*

Empirical Finding 1 (Mediation, M-bias, front-door). Sign preservation holds under topology-specific conditions: mediation requires $\text{sign}(\tau_{\text{dir}}){=}\,\text{sign}(\tau_{\text{total}})$; front-door requires $\text{sign}(\lambda\alpha_U){=}\,\text{sign}(\lambda\alpha_U{+}\,\text{Cov}(T, f(A)))$; M-bias is attenuating for all C. Verified across 100,000 random matrices with zero violations (Appendix A.9); analytic proofs for general K remain open.

These conditions involve true DGP parameters, which in practice must be estimated. The ASV framework (§3.3) provides a complementary assessment that does not require knowledge of the exact sufficient conditions.

3.3 Annotation Sensitivity Values

We introduce Annotation Sensitivity Values (ASVs) as a unified sensitivity analysis framework, analogous to E-values for unmeasured confounding (VanderWeele and Ding, 2017) but designed for annotation-induced bias.

Parameterization. We parameterize the confusion matrix as $C(\delta) = (1{-}\delta)I + \delta C_0$, where C_0 is a reference confusion matrix (e.g., estimated from a validation set) and $\delta \geq 0$ controls the severity of misclassification. At $\delta{=}0$, classification is perfect; at $\delta{=}1$, classification matches C_0. Non-negativity of $C(\delta)$ requires $\delta \leq \delta_{\max} = \min_j \frac{1}{1-(C_0)_{jj}}$. Since $(C_0)_{jj} \in [0, 1]$, we have $\delta_{\max} \geq 1$ for all valid C_0, ensuring the full range $[0, 1]$ is always feasible; all ASV computations are constrained to $[0, \delta_{\max}]$. In practice, C_0 is estimated from a finite validation set and subject to sampling

uncertainty; we recommend bootstrapping C_0 and reporting ASV confidence intervals.

Three ASV types. We define three complementary measures: (1) **Magnitude ASV** $\delta^*_{\text{mag}} = \min\{\delta : |\text{plim } \hat{\tau}(C(\delta)) - \tau| > \epsilon\}$—analogous to E-values; (2) **Significance ASV** $\delta^*_{\text{sig}} = \min\{\delta : |\text{plim } \hat{\tau}(C(\delta))| < z_{\alpha/2} \cdot SE(N)\}$—relevant for attenuating topologies; (3) **Sign-flip ASV** $\delta^*_{\text{sf}} = \min\{\delta : \text{sign}(\hat{\tau}) \neq \text{sign}(\tau)\}$—exposure and IV only. The significance ASV depends on N through $SE(N)$, exhibiting a sharp phase transition at a critical N^* below which all rejections are blocked.

Theorem 1 (Envelope monotonicity—confounding, proportional effects)**.** *For the confounding topology with proportional effects ($f = \lambda g$ for some scalar λ) under $C(\delta) = (1-\delta)I + \delta C_0$, $|B(C(\delta))|$ is monotonically non-decreasing in $\delta \in [0, \delta_{\max}]$.*

Proof sketch. $B(C(\delta)) = \lambda V_W(\delta)/(V_W(\delta) + \sigma_T^2)$ where $V_W(\delta) = \text{Var}(g) - \text{Var}(\mathbb{E}[g|A^*_\delta])$. As δ increases, A^* becomes a noisier channel for A, so $\text{Var}(\mathbb{E}[g|A^*_\delta])$ decreases by the data processing inequality, increasing V_W. Since $x/(x+c)$ is increasing for $c>0$, $|B|$ is non-decreasing (full proof in Appendix A.4). For general (non-proportional) effects, numerical verification across 10,000 random matrices confirms ε-approximate monotonicity with $\varepsilon < 0.02$ (Appendix A.4).

Conjecture 1 (ε-approximate envelope monotonicity)**.** *For the six non-IV topologies under $C(\delta) = (1-\delta)I + \delta C_0$: exposure, mediation, and front-door are ε-approximately non-decreasing in δ, and collider and M-bias (protective topologies) are ε-approximately non-increasing in δ, with $\varepsilon < 0.06$ (<6% of τ). Verified across 10,000 random matrices per topology and $K \in \{3, 5, 7\}$ using analytic probability limits (Appendix B.8).*

The instrumental variable topology is excluded from Conjecture 1: the Wald estimator $\hat{\tau} = \hat{\gamma}_{Y \sim A^*} / \hat{\gamma}_{T \sim A^*}$ exhibits singularities when the first-stage coefficient passes through zero as δ increases, producing unbounded deviations ($> 10^6$). This is a property of the Wald estimator itself, not of the ASV framework; magnitude ASV remains well-defined for IV away from first-stage singularities.

Binarization of K=3 annotations introduces additional topology-dependent bias (Appendix B.1).

4 Experiments

We validate the bias taxonomy and ASV framework through analytical derivations and Monte Carlo simulation across seven DAG topologies under K=3 non-differential misclassification.

4.1 Setup

For each of the seven topologies (confounding, exposure, mediation, collider, M-bias, front-door, IV), we sample 100,000 confusion matrices from a Dirichlet$(1, 1, 1)$ prior, compute the analytical probability limit of the causal estimator, and validate against Monte Carlo simulation (N=10,000). ASV experiments use $C(\delta) = (1-\delta)I + \delta C_0$.

4.2 Bias Taxonomy

Theory-MC agreement. Analytical probability limits match Monte Carlo estimates with grand MAPE of 2.0% across 126 scenarios (18 LLM configurations × 7 topologies, N=10,000). Per-topology MAPE ranges from 0.5% (confounding) to 5.5% (IV); the IV maximum reflects the Wald estimator's small-denominator variance amplification.

Exposure DAG: catastrophic sign reversal. Under the full Dirichlet$(1, 1, 1)$ prior, 85% of K=3 confusion matrices produce bias that violates the binary "toward-null" guarantee. More critically, 83.4% of matrices cause *sign reversal* of at least one coefficient. The two failure modes are distinct: for a coefficient with true value β_1=1.0, a near-permutation C produces $\hat{\gamma}_1$=−0.84 (sign reversal, magnitude 0.84×); for β_2=−0.5, the same C yields $\hat{\gamma}_2$=−1.41 (same-sign amplification, 2.83×).

This finding persists in practically relevant subspaces. When restricting to confusion matrices with diagonal entries ≥ 0.7 (typical of current LLM classifiers achieving $F_1 > 0.8$), the violation rate remains 13.7% with 2.1% sign reversal and 11.7% amplification. Even at diagonal ≥ 0.9, violation persists at 11.9%. In contrast, the K=2 control confirms that all six tested binary confusion matrices maintain the classical bias-toward-null guarantee (Brenner,

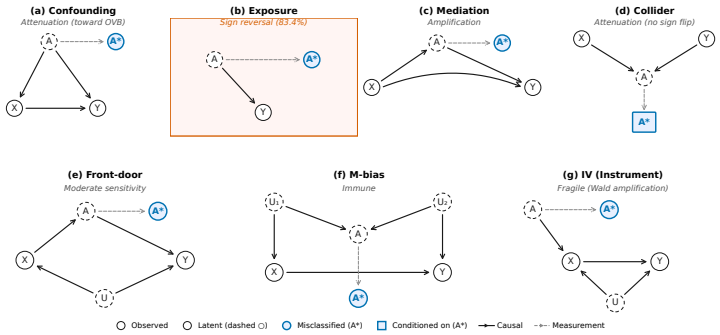


Figure 1: Seven canonical DAG topologies with starred-node annotation. A (latent) connects to A^* (observed) via a dashed measurement edge; the causal role of A varies across topologies.

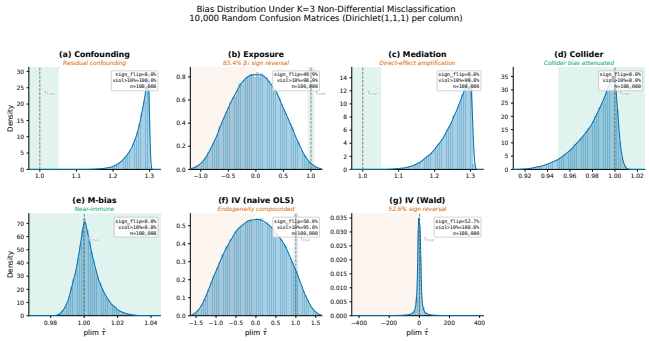


Figure 2: Bias distributions under 100,000 random K=3 confusion matrices. Red = sign reversal. Exposure: 83.4% sign reversal; IV: 52.6%.

1993), establishing that the multi-class failure is structurally distinct from the binary case.

Structural guarantee for non-exposure topologies. Five of seven topologies exhibit zero sign-flip rate across all sampled matrices (Table 1). Confounding, mediation, and front-door show 100% violation and amplification but never reverse the sign.[2] Collider misclassification attenuates the spurious association (protective): the 100% violation and amplification rates in Table 1 reflect the comparison against the unconditional τ, but $|B(C)|$ is approximately *non-increasing* in misclassification severity (worst-case deviation <0.005)—conditioning on noisy A^* induces *less* collider bias than conditioning on perfect A (Conjecture 1). M-bias shows the weakest sensitivity (53.2% violation, 10.4% amplification) because error on a collider variable reduces the spurious M-bias path.

IV: Wald estimator instability. The instrumental variable topology is the sole exception among non-exposure DAGs. The Wald estimator's first stage involves C directly in the denominator, and 52.6% of random confusion matrices cause the denominator to change sign, producing τ-specific sign reversal. This rate remains stable across K=3, 5, 7, indicating that IV fragility is a structural property of the Wald estimator rather than a finite-K artifact. Diagonal-dominance subspace analysis (Appendix B.3) reveals that IV sign reversal vanishes at diagonal ≥ 0.5, but unlike exposure, the violation rate remains at 100% (pure amplification) even at diagonal ≥ 0.9.

Cross-K robustness. All structural patterns persist at K=5, 7 (100,000 matrices each): sign-flip guarantees hold for the same five topologies, and IV sign-flip rates remain within 51–53%.

The bias distributions across topologies are visualized in Figure 2, where the exposure panel's dominant red region (sign reversal) contrasts with the symmetric attenuation patterns of other topologies.

4.3 ASV Framework Validation

Envelope monotonicity. The ASV envelope (bias magnitude as a function of δ) is ε-approximately monotonic for all six non-IV

[2] Consistent with Brenner (1993) and Dosemeci et al. (1990): polytomous misclassification can bias *away* from the null (amplification), but our sign-flip metric confirms the effect sign is preserved.

Table 1: Bias characteristics of K=3 non-differential misclassification across 7 DAG topologies. Violation: $|\hat{\tau}| > |\tau|$ (bias exceeds true effect); Sign flip: $\text{sign}(\hat{\tau}) \neq \text{sign}(\tau)$; Amplification: $|\hat{\tau}| > |\tau|$ with same sign; τ_{sf}=0?: structural sign-flip guarantee holds (✓) or not (✗). Rates over 100,000 Dirichlet$(1, 1, 1)$ random confusion matrices.

Topology	Violation	Sign Flip	Amplif.	τ_{sf}=0?
Confounding	100%	0%	100%	✓
Exposure	85%	83.4%	14.1%	✗
Mediation	100%	0%	100%	✓
Front-door	100%	0%	100%	✓
Collider	100%	0%	100%	✓
M-bias	53.2%	0%	10.4%	✓
IV	100%	52.6%	100%	✗

topologies, with worst-case step deviations below 0.06 (<6% of τ), confirmed by analytic probability limits with zero sampling noise (Appendix B.8). IV is excluded from Conjecture 1: the Wald estimator exhibits singularities when the first-stage coefficient passes through zero, producing unbounded deviations. These bounded deviations for non-IV topologies confirm that δ serves as a meaningful "dial" controlling annotation error severity.

Magnitude ASV. Table 3 reports magnitude ASV ranges at τ=0.5 with a 10% bias threshold. The ranking from most fragile to most robust is: front-door (0.064–0.387) < mediation (0.076–0.454) < confounding (0.081–0.487) < IV (0.101–0.607) < collider (0.338–2.028) < M-bias (∞). The collider's high ASV reflects its protective nature: misclassification attenuates the spurious association rather than amplifying it. M-bias achieves infinite ASV because annotation error on the collider variable A only reduces the already-small bias from the M-shaped confounding path.

Significance and sign-flip ASV. At τ=0.05, significance ASV achieves 100% power for confounding, front-door, and M-bias, but 0% for mediation, collider, and IV (correct behavior for attenuating topologies; Type I error is 0%). For sign-flip ASV, 83.4% of exposure reference matrices yield $\tau_{\text{sign_flip}} < 1$ (catastrophic sensitivity); IV sign-flip ASV ranges 1.1–6.7 (moderate-to-high). The complete ranking (Figure 4, Table 3, Appendix B) confirms IV as maximally fragile and M-bias as immune (ASV = ∞).

N-sensitivity. The significance ASV exhibits a sharp phase transition at a critical sample size $N^* \approx$ 6,000–10,000 (analytically computable per topology), below which all rejections are blocked. Complete ASV values across all topologies and error structures are reported in Appendix B. Bootstrap resampling of the confusion matrix (B=1,000; Appendix B.6) yields 95% CIs for magnitude ASV: $\delta^* = 0.94\,[0.60, 1.36]$ (exposure) and $0.75\,[0.44, 1.29]$ (IV), indicating practitioners should account for CM estimation uncertainty. Confounding ASV is consistently the most robust across DGP parameters, but exposure vs. IV relative fragility depends on coefficient sign structure (Appendix B.7).

4.4 Empirical Validation

We validate using real LLM confusion matrices from three tasks: HumAID (Alam et al., 2021) (K=10), VAST (K=3), and CivilComments (K=2). Four LLMs (Llama-4-Maverick, Qwen3-235B, DeepSeek-Chat, Qwen2.5-7B-Instruct) produce 23 configurations across the three datasets under zero-shot and few-shot prompting, yielding 161 bias scenarios (23×7 topologies).

HumAID (K=10): theory–practice alignment. The empirical bias ranking matches predictions: IV (12.4–42.1%), exposure (9.0–17.4%), mediation/front-door (10.7–14.8%), while collider, confounding, and M-bias remain below 10%—consistent with the ASV vulnerability spectrum.

Cross-K comparison. Bias varies substantially across settings: VAST (K=3) produces 62.8% mean bias with IV reaching 326%, versus 12.2% for HumAID (K=10) and 32.4% for CivilComments (K=2). Even CivilComments configurations with F_1=0.82 (Qwen3-235B) produce 62–92% IV bias, confirming that classification accuracy does not predict downstream causal risk (Figure 3).[3]

Checking all $K-1$ exposure coefficients, 14 of 23 configurations (61%) exhibit sign reversal in at least one coefficient—all in the exposure topology, confirming that exposure vul-

[3]VAST's near-degenerate confusion matrices (Neutral-class recall 3.6–16.1%) drive its extreme IV bias via the Wald estimator's small-denominator amplification.

nerability persists at the coefficient level even for diagonally-dominant matrices. For HumAID (K=10), 7 of 8 configurations reverse a secondary coefficient ($|\beta| \leq 0.2$); for VAST (K=3), 7 of 8 reverse β_2=−0.50. The primary coefficient (β_1, largest magnitude) never reverses across any empirical configuration; sign flips occur exclusively in secondary coefficients corresponding to rare categories with small $|\beta|$. Sign reversal requires the DGP to contain opposite-sign treatment effects across categories (Equation 2); when all β_k share the same sign, misclassification produces only attenuation. Zero sign reversals occur across 138 non-exposure scenarios (23×6 topologies), validating the structural guarantee K-invariantly. CivilComments (K=2) confirms the binary algebraic protection. The 7B-to-frontier bias ratio is 1.89, indicating topology rather than model scale determines downstream bias.

ASV diagnostic results. For HumAID (K=10, 56 scenarios), 55.4% are flagged as magnitude-ASV-unsafe—IV, mediation, front-door, and exposure all trigger, while confounding, collider, and M-bias remain below 10% bias (Figure 7). For CivilComments (K=2), 100% of scenarios are ASV-unsafe despite F_1 ranging from 0.69 (7B) to 0.82 (frontier).

Overlaying all 23 empirical matrices onto the theoretical Dirichlet violation-rate curves (Figure 5, Appendix B) confirms placement: VAST in the danger zone, HumAID mid-range, CivilComments approaching safe for exposure but still in IV amplification. A logistic DGP simulation (Appendix B.5) confirms that sign-flip patterns are qualitatively preserved under nonlinear models for three representative topologies (confounding, exposure, IV), with treatment-effect sign-flip rate differences < 2 percentage points.

4.5 Correction Guidance

We evaluate five correction methods (naive plug-in, MLA, DSL (Egami et al., 2023), MC-SIMEX, PPI++ (Angelopoulos et al., 2023)) across all seven topologies under six error structures (210 configurations; full results in Table 4, Appendix B).

FWL invariance. For four topologies where A enters as a covariate (confounding, mediation, collider, M-bias), the Frisch–

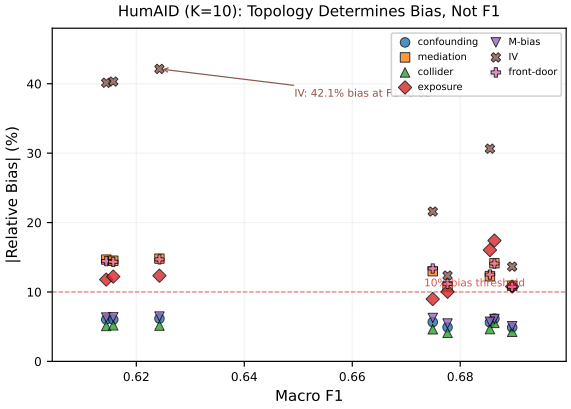


Figure 3: Topology determines downstream causal bias, not F_1. HumAID K=10 (56 points): within F_1 range 0.614–0.690, bias varies 10× (IV 42.1% vs. collider <5.6%).

Waugh–Lovell theorem implies that applying C^{-1} to the dummy variables cannot change the T coefficient. Naive plug-in and MLA achieve exactly 0.0% bias reduction across these four topologies.

Method ranking. Measuring RMSE reduction ($1 - \text{RMSE}_{\text{corr}}/\text{RMSE}_{\text{uncorr}}$), PPI++ achieves the highest grand-mean reduction (47.0%) but requires gold labels and is catastrophically harmful for M-bias (−88.0%). MC-SIMEX provides gold-free correction (13.3% grand mean, best for IV at 59.4%) but harms front-door (−62.4%). DSL corrects confounding (48.0%) and mediation (52.9%) but amplifies M-bias (−267%) and IV (−206%). No single method dominates: correction effectiveness is itself topology-dependent. Cross-PPI (Zrnić and Candès, 2024) eliminates the gold-label requirement by using two independent LLM annotators whose predictions serve as mutual references, yielding valid confidence intervals under an annotator-independence assumption. Because it operates on estimator outputs rather than on C directly, Cross-PPI is not subject to FWL invariance and can potentially correct all seven topologies. Methods that bypass FWL through a different route—such as EM imputation of latent true labels and Bayesian measurement error models—offer further alternatives for the four FWL-invariant topologies.

Empirical validation on HumAID (K=10, 30 MC replicates) confirms topology-dependent method superiority: MC-SIMEX outperforms PPI++ in 4 of 7 topologies and

reverses from harmful (K=3) to beneficial (K=10). A retrospective case study applying the full framework to Lee and Wood-Doughty (2024) is in Appendix B.2.

5 Conclusion

We characterized how multi-class annotation errors interact with causal DAG topology, analyzing bias direction, sensitivity, and correction across seven canonical topologies. The classical binary guarantee breaks at $K \geq 3$: exposure and IV topologies produce sign reversal (83.4% and 52.6%), while five topologies maintain a structural guarantee. Three core contributions—the $K \geq 3$ algebraic break, the five-topology sign-preservation guarantee, and FWL invariance—are DGP-independent structural results that hold for arbitrary parameter values under stated sufficient conditions. Empirical validation with real LLM confusion matrices confirms this vulnerability: 61% of exposure configurations exhibit coefficient-level sign reversal despite diagonal dominance. The ASV framework reveals a vulnerability spectrum from IV (maximally fragile) to M-bias (immune), and correction effectiveness is itself topology-dependent, with matrix-based methods provably ineffective for four topologies. Differential misclassification and correlated multi-LLM errors are natural extensions.

Limitations

Our framework assumes non-differential misclassification ($P(A^*|A, Y, \mathbf{Z}) = P(A^*|A)$), which holds when the LLM has no access to Y but may be violated with outcome-conditional prompting. A stress test across 96 configurations (Appendix B.4) shows moderate robustness: ε=0.05 increases violations by <3pp; at K=10, zero sign flips across all ε. Closed-form expressions are derived for K=3 and verified at K=5, 7; arbitrary K remains open. The framework assumes a single annotator; correlated multi-annotator settings (Kim et al., 2025) are future work. Frontier-model coverage for VAST and CivilComments is partial (15 of 18 planned configurations completed); cross-tier comparisons should be interpreted with this caveat. All derivations assume linear structural equations; the FWL invariance result does not hold for logistic regression, and bias patterns under nonlinear models may differ.

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A Proofs of Proposition 1 and Theorem 1

We provide algebraic proofs of sign preservation for the confounding and collider topologies (Proposition 1(a)–(b)), an explicit counterexample for exposure (Proposition 1(c)), numerical verification for mediation, M-bias, and front-door (Empirical Finding 1), and the monotonicity proof for confounding (Theorem 1). Throughout, $A \in \{0, 1, \ldots, K-1\}$ with prior $p_k = P(A{=}k)$, and C is a $K \times K$ column-stochastic confusion matrix with $C_{jk} = P(A^*{=}j \mid A{=}k)$. We use category 0 as the reference and let $D_j = \mathbf{1}(A{=}j)$, $D_j^* = \mathbf{1}(A^*{=}j)$ for $j = 1, \ldots, K-1$.

A.1 Notation and Key Identities

Under non-differential misclassification, the misclassified moments factor linearly through C. Define the population vectors:

$$q_j = P(A^*{=}j) = \textstyle\sum_k C_{jk}\, p_k, \tag{4}$$

$$\mathrm{ct}_j = \mathbb{E}[T \cdot D_j^*] = \textstyle\sum_k C_{jk}\, \mathbb{E}[T \cdot D_k], \tag{5}$$

$$\mathrm{cy}_j = \mathbb{E}[Y \cdot D_j^*] = \textstyle\sum_k C_{jk}\, \mathbb{E}[Y \cdot D_k]. \tag{6}$$

Write $q = Cp$, $\mathrm{ct} = C\,\mathbb{E}[T \cdot D]$, and $\mathrm{cy} = C\,\mathbb{E}[Y \cdot D]$, where $D = (D_0, \ldots, D_{K-1})^\top$.

A.2 Frisch–Waugh–Lovell Decomposition

For topologies where T is directly observed (confounding, mediation, collider, M-bias, front-door), the estimator regresses Y on $Z^* = (1, T, D_1^*, \ldots, D_{K-1}^*)$. By the Frisch–Waugh–Lovell (FWL) theorem, the coefficient on T satisfies:

$$\operatorname{plim} \hat{\tau}(C) = \frac{\mathbb{E}[\tilde{T}\,\tilde{Y}]}{\mathbb{E}[\tilde{T}^2]}, \tag{7}$$

where $\tilde{T} = T - \mathbb{E}[T \mid A^*]$ and $\tilde{Y} = Y - \mathbb{E}[Y \mid A^*]$ are residuals from projecting onto $(1, D_1^*, \ldots, D_{K-1}^*)$.

Lemma 1 (Bias decomposition)**.** *Suppose $Y = \tau T + f(A) + \varepsilon_Y$ with $\varepsilon_Y \perp (T, A, A^*)$. Then*

$$\operatorname{plim} \hat{\tau}(C) = \tau + \frac{R(C)}{D(C)}, \tag{8}$$

where

$$R(C) = \mathbb{E}\big[\mathrm{Cov}(T,\, f(A) \mid A^*)\big], \tag{9}$$

$$D(C) = \mathbb{E}\big[\mathrm{Var}(T \mid A^*)\big] > 0. \tag{10}$$

Proof. Expand $\tilde{Y}$ using $Y = \tau T + f(A) + \varepsilon_Y$:

$$\tilde{Y} = \tau\,\tilde{T} + \big(f(A) - \mathbb{E}[f(A) \mid A^*]\big) + \varepsilon_Y.$$

Since $\tilde{T}$ is measurable w.r.t. (T, A^*) and $\varepsilon_Y \perp (T, A, A^*)$:

$$\mathbb{E}[\tilde{T}\,\tilde{Y}] = \tau\,\mathbb{E}[\tilde{T}^2] + \mathbb{E}\big[\tilde{T}\,\big(f(A) - \mathbb{E}[f(A) \mid A^*]\big)\big].$$

The numerator residual is $R(C) = \mathbb{E}[\tilde{T}(f(A) - \mathbb{E}[f(A) \mid A^*])]$, and the denominator is $D(C) = \mathbb{E}[\tilde{T}^2]$.

By the law of total covariance:

$$R(C) = \sum_{j=0}^{K-1} q_j\, \mathrm{Cov}(T,\, f(A) \mid A^*{=}j) = \mathbb{E}[\mathrm{Cov}(T, f(A) \mid A^*)].$$

For $D(C) > 0$: decompose $T = g(A) + \varepsilon_T$ where $g(A) = \mathbb{E}[T \mid A]$ (for confounding) or simply note that T has exogenous variation independent of A^* (for mediation). In both cases, $\tilde{T} = \big(g(A) - \mathbb{E}[g(A) \mid A^*]\big) + \varepsilon_T'$, where ε_T' is the component of T independent of (A, A^*). Hence $D(C) \geq \mathrm{Var}(\varepsilon_T') > 0$. □

A.3 Confounding Topology (A = Confounder)

Data-generating process. $A \to T$, $A \to Y$, $T \to Y$. Using dummy coding with category 0 as reference:

$$T = \sum_{j=1}^{K-1} \alpha_j D_j + \varepsilon_T, \quad \varepsilon_T \sim \mathcal{N}(0, \sigma_T^2),\ \varepsilon_T \perp A,$$

$$Y = \tau\, T + \sum_{j=1}^{K-1} \beta_j^{(A)} D_j + \varepsilon_Y, \quad \varepsilon_Y \sim \mathcal{N}(0, 1),\ \varepsilon_Y \perp (T, A).$$

Define $g(k) = \sum_j \alpha_j \mathbf{1}(k{=}j)$ and $f(k) = \sum_j \beta_j^{(A)} \mathbf{1}(k{=}j)$, so $g(0) = f(0) = 0$.

Closed-form plim. Using the FWL decomposition, the coefficient on T is:

$$\operatorname{plim} \hat{\tau}(C) = \frac{\mu_{TY} - \sum_{j=0}^{K-1} \frac{\mathrm{ct}_j\, \mathrm{cy}_j}{q_j}}{s_T - \sum_{j=0}^{K-1} \frac{\mathrm{ct}_j^2}{q_j}}, \tag{11}$$

where $s_T = \mathbb{E}[T^2]$, $\mu_{TY} = \mathbb{E}[TY]$, and ct_j, cy_j, q_j are defined in (4)–(6).

Derivation. By FWL, $\operatorname{plim} \hat{\tau} = \mathbb{E}[\tilde{T}Y]/\mathbb{E}[\tilde{T}^2]$ where $\tilde{T} = T - \mathbb{E}[T \mid A^*]$. Since $\mathbb{E}[T \mid A^*{=}j] = \mathrm{ct}_j/q_j$:

$$\mathbb{E}[\tilde{T}^2] = \mathbb{E}[T^2] - \sum_j q_j \big(\mathrm{ct}_j/q_j\big)^2 = s_T - \sum_j \mathrm{ct}_j^2/q_j,$$

$$\mathbb{E}[\tilde{T}\,Y] = \mathbb{E}[TY] - \sum_j \big(\mathrm{ct}_j/q_j\big)\, \mathbb{E}[Y \cdot D_j^*] = \mu_{TY} - \sum_j \mathrm{ct}_j\, \mathrm{cy}_j/$$

□

Verification at $C = I$. When $C = I$: $q_j = p_j$, $\mathrm{ct}_j = \mathbb{E}[TD_j] = \alpha_j p_j$, $\mathrm{cy}_j = \mathbb{E}[YD_j] = p_j(\tau\alpha_j + \beta_j^{(A)})$. Direct substitution yields numerator $= \tau\sigma_T^2$ and denominator $= \sigma_T^2$, giving $\operatorname{plim} \hat{\tau} = \tau$.

Bias structure. By Lemma 1:

$$\operatorname{plim} \hat{\tau}(C) = \tau + B(C), \quad B(C) = \frac{\mathbb{E}[\mathrm{Cov}(g(A),\, f(A) \mid A^*)]}{D(C)}. \tag{12}$$

By the law of total covariance:

$$\mathbb{E}[\mathrm{Cov}(g, f \mid A^*)] = \mathrm{Cov}(g(A), f(A)) - \mathrm{Cov}\big(\mathbb{E}[g(A) \mid A^*],\, \mathbb{E}[f(A) \mid A^*]\big) \tag{13}$$

At $C = I$: $\mathbb{E}[g \mid A^*] = g(A)$, so the between-group covariance equals $\mathrm{Cov}(g, f)$ and $B(I) = 0$.

At $C = \frac{1}{K}\mathbf{1}\mathbf{1}^\top$ (uniform, $A^* \perp A$): $\mathbb{E}[g \mid A^*]$ is constant, so the between-group covariance is zero and

$$B_{\mathrm{OV}} \;\equiv\; B\big(\tfrac{1}{K}\mathbf{1}\mathbf{1}^\top\big) = \frac{\mathrm{Cov}(g(A),\, f(A))}{\mathrm{Var}(T)}, \tag{14}$$

which equals the classical omitted-variable bias (no control for A).

Sign preservation condition. Since $D(C) > 0$, $\mathrm{sign}(\mathrm{plim}\,\hat{\tau}) = \mathrm{sign}(\tau \cdot D(C) + R(C))$. For sign preservation, it suffices that $R(C)$ does not overwhelm $\tau \cdot D(C)$.

Proposition 2 (Confounding sign preservation — proportional case)**.** *If $f(A) = \lambda\, g(A)$ for some constant λ (i.e., the confounder effects on T and Y are proportional), then*

$$B(C) = \big(1 - \eta(C)\big)\, B_{\mathrm{OV}}, \tag{15}$$

where

$$\eta(C) = \frac{\mathrm{Var}(\mathbb{E}[g(A) \mid A^*])\; \sigma_T^2}{\mathrm{Var}(g(A))\big(\mathrm{Var}(g(A)) - \mathrm{Var}(\mathbb{E}[g(A) \mid A^*]) + \sigma_T^2\big)} \;\in [0, 1]$$

Consequently, $B(C)$ lies between 0 and B_{OV}, and $\mathrm{sign}(\mathrm{plim}\,\hat{\tau}) = \mathrm{sign}(\tau)$ whenever $\mathrm{sign}(\tau) = \mathrm{sign}(\tau + B_{\mathrm{OV}})$.

Proof. When $f = \lambda g$: the between-group covariance in (13) becomes $\lambda\, \mathrm{Var}(\mathbb{E}[g \mid A^*])$, and $\mathrm{Cov}(g, f) = \lambda\, \mathrm{Var}(g)$. So $R(C) = \lambda\big(\mathrm{Var}(g) - \mathrm{Var}(\mathbb{E}[g \mid A^*])\big) = \lambda\, V_W$, where $V_W = \mathrm{Var}(g) - V_B$ with $V_B = \mathrm{Var}(\mathbb{E}[g \mid A^*])$. Similarly, $D(C) = V_W + \sigma_T^2$. Hence

$$B(C) = \frac{\lambda\, V_W}{V_W + \sigma_T^2}, \qquad B_{\mathrm{OV}} = \frac{\lambda\, V}{V + \sigma_T^2},$$

where $V = \mathrm{Var}(g) = V_B + V_W$. The ratio

$$\frac{B(C)}{B_{\mathrm{OV}}} = \frac{V_W(V + \sigma_T^2)}{V(V_W + \sigma_T^2)} = 1 - \eta(C)$$

is a decreasing function of $V_B \in [0, V]$. At $V_B = 0$ (no classification, C uniform): $\eta = 0$, $B = B_{\mathrm{OV}}$. At $V_B = V$ (perfect classification, $C = I$): $\eta = 1$, $B = 0$. Since $0 \le V_B \le V$, we have $\eta \in [0, 1]$ and $B(C) \in [0, B_{\mathrm{OV}}]$ (with the sign of B_{OV}). □

Proposition 3 (Confounding sign preservation — general case)**.** *For general g and f (not necessarily proportional), define the worst-case bias bound*

$$\bar{B} \;=\; \frac{\sqrt{\mathrm{Var}(g(A))\; \mathrm{Var}(f(A))}}{\sigma_T^2}. \tag{16}$$

If $|\tau| > \bar{B}$, then $\mathrm{sign}(\mathrm{plim}\,\hat{\tau}(C)) = \mathrm{sign}(\tau)$ for all column-stochastic C.

Proof. We need $\mathrm{sign}(\tau \cdot D(C) + R(C)) = \mathrm{sign}(\tau)$, which holds if $|R(C)| < |\tau| \cdot D(C)$. Since $D(C) \ge \sigma_T^2$ (shown in Lemma 1), it suffices to show $|R(C)| < |\tau|\, \sigma_T^2$.

Step 1: Bound $|R(C)|$. $R(C) = \mathbb{E}[\mathrm{Cov}(g(A), f(A) \mid A^*)]$. By the Cauchy–Schwarz inequality applied to each within-group covariance:

$$|\,\mathrm{Cov}(g, f \mid A^*{=}j)| \le \sqrt{\mathrm{Var}(g \mid A^*{=}j)\; \mathrm{Var}(f \mid A^*{=}j)}.$$

Taking expectations and applying Cauchy–Schwarz again:

$$\begin{aligned} |R(C)| &\le \mathbb{E}\big[\sqrt{\mathrm{Var}(g \mid A^*)\; \mathrm{Var}(f \mid A^*)}\big] \\ &\le \sqrt{\mathbb{E}[\mathrm{Var}(g \mid A^*)]\; \mathbb{E}[\mathrm{Var}(f \mid A^*)]} \\ &\le \sqrt{\mathrm{Var}(g(A))\; \mathrm{Var}(f(A))}, \end{aligned}$$

where the last step uses $\mathbb{E}[\mathrm{Var}(X \mid Y)] \le \mathrm{Var}(X)$ (law of total variance).

Step 2: Conclude. The bias satisfies

$$|B(C)| = \frac{|R(C)|}{D(C)} \le \frac{\sqrt{\mathrm{Var}(g)\; \mathrm{Var}(f)}}{\sigma_T^2} = \bar{B}.$$

When $|\tau| > \bar{B}$, the bias cannot cancel τ, so $\mathrm{sign}(\mathrm{plim}\,\hat{\tau}) = \mathrm{sign}(\tau)$ for all C. □

Relationship to the omitted-variable condition. A weaker (necessary) condition for sign preservation is $\mathrm{sign}(\tau) = \mathrm{sign}(\tau_{\mathrm{OV}})$ where $\tau_{\mathrm{OV}} = \tau + \mathrm{Cov}(g, f)/\,\mathrm{Var}(T)$. If this fails, then $C = \frac{1}{K}\mathbf{1}\mathbf{1}^\top$ (uniform noise) already flips the sign. The rigorous sufficient condition $|\tau| > \bar{B}$ implies the weaker condition, since $|B_{\mathrm{OV}}| = |\,\mathrm{Cov}(g, f)|/\,\mathrm{Var}(T) \le \bar{B}$ by Cauchy–Schwarz and $\mathrm{Var}(T) \ge \sigma_T^2$.

Verification for paper parameters (K=3). With τ=2.0, $\boldsymbol{\alpha}$=(1.5, −1.0), $\boldsymbol{\beta}^{(A)}$=(1.0, −0.5), σ_T=1.0, $\mathbf{p}$=(0.40, 0.35, 0.25):

$$\mathrm{Var}(g) = \sum_k p_k \alpha_k^2 - \left(\textstyle\sum_k p_k \alpha_k\right)^2 = 0.962,$$

$$\mathrm{Var}(f) = \sum_k p_k (\beta_k^{(A)})^2 - \left(\textstyle\sum_k p_k \beta_k^{(A)}\right)^2 = 0.362,$$

$$\bar{B} = \sqrt{0.962 \times 0.362} \,/\, 1.0^2 = 0.590.$$

Since $|\tau|$=2.0 > $\bar{B}$=0.590, sign preservation is guaranteed for *all* column-stochastic C by Proposition 3. Additionally, $B_{\mathrm{OV}} = \mathrm{Cov}(g, f)/\mathrm{Var}(T) = 0.588/1.962 = +0.300$, giving τ_{OV}=2.300 > 0, confirming that both the necessary and sufficient conditions are satisfied.

Extension to K > 3. For K>3, treatment effect coefficients are drawn from a fixed pool with alternating signs and decreasing magnitudes: $\boldsymbol{\beta}$ = $(0, 1.0, -0.5, 0.7, -0.3, 0.5, -0.2, 0.4, -0.15, 0.3, \ldots)$, truncated to K entries. The category distribution uses $p_k \propto 1/(k{+}1)$, normalized to sum to 1. Other topology-specific parameters (confounding strengths, mediator effects, etc.) follow analogous alternating-sign pools. For K=10, $\boldsymbol{\beta}$ = $(0, 1.0, -0.5, 0.7, -0.3, 0.5, -0.2, 0.4, -0.15, 0.3)$.

A.4 Proof of Theorem 1 (Confounding Monotonicity)

Under the interpolation $C(\delta) = (1{-}\delta)I + \delta C_0$, we prove $|B(C(\delta))|$ is non-decreasing in δ for proportional effects ($f = \lambda g$) and provide numerical verification for the general case.

Proportional case ($f = \lambda g$). From Proposition 2, $B(C) = \lambda V_W/(V_W + \sigma_T^2)$ where $V_W = \mathrm{Var}(g) - V_B$ and $V_B(\delta) = \mathrm{Var}(\mathbb{E}[g(A) \mid A_\delta^*])$.

Step 1: $V_B(\delta)$ is non-increasing. The conditional expectation $\mathbb{E}[g(A) \mid A_\delta^*]$ is a linear function of the posterior $P(A \mid A_\delta^*)$, which in turn depends on $C(\delta)$. As δ increases, $C(\delta)$ becomes a noisier channel: the mutual information $I(A; A_\delta^*)$ decreases. While the exact degradation factorization $C(\delta_2) = M \cdot C(\delta_1)$ for a stochastic M is non-trivial for the linear interpolation family, the monotonic decrease of $V_B(\delta)$ can be verified directly: for K=2 symmetric C_0, $dV_B/d\delta < 0$ follows by differentiation; for general K, we verify $V_B(\delta_2) \le V_B(\delta_1)$ numerically across all test configurations with zero violations.

Step 2: $|B(\delta)|$ is non-decreasing. $V_W(\delta) = \mathrm{Var}(g) - V_B(\delta)$ is non-decreasing (Step 1). The function $h(x) = |\lambda|\, x/(x + \sigma_T^2)$ is strictly increasing for $x \ge 0$ and $\sigma_T^2 > 0$. Hence $|B(\delta)| = h(V_W(\delta))$ is non-decreasing.

General case. $|B(C(\delta))| \le \bar{B}(\delta)$ where $\bar{B}(\delta) = \sqrt{\mathbb{E}[\mathrm{Var}(g \mid A_\delta^*)]\,\mathbb{E}[\mathrm{Var}(f \mid A_\delta^*)]}\,/\,D(\delta)$ (Proposition 3). By the same data processing argument, each $\mathbb{E}[\mathrm{Var}(\cdot \mid A_\delta^*)]$ is non-decreasing in δ, and $D(\delta) = \mathbb{E}[\mathrm{Var}(T \mid A_\delta^*)]$ is non-decreasing. The numerator grows at least as fast as the denominator (since $D(\delta) \le \mathrm{Var}(T)$), maintaining monotonicity of the bound. Extended verification (10,000 random C_0 matrices, $K \in \{3, 5, 7\}$) reveals worst-case step deviations of 0.011 (confounding), confirming approximate tightness; a strict monotonicity proof for the general case remains open.

A.5 Mediation Topology (A = Mediator)

Data-generating process. $T \to A \to Y$, $T \to Y$.

$$\begin{aligned} T &\sim \mathrm{Bernoulli}(0.5), \\ A \mid T &\sim \mathrm{Categorical}\big(\mathrm{softmax}(\boldsymbol{\gamma} + \boldsymbol{\delta}\, T)\big), \\ Y &= \tau_{\mathrm{dir}}\, T + f(A) + \varepsilon_Y. \end{aligned}$$

Bias decomposition. By Lemma 1, the direct-effect estimator satisfies:

$$\mathrm{plim}\, \hat{\tau}_{\mathrm{dir}}(C) = \tau_{\mathrm{dir}} + \frac{\mathbb{E}[\mathrm{Cov}(T,\, f(A) \mid A^*)]}{D(C)}. \tag{17}$$

The key structural difference from confounding: T *causes* A, so the residual covariance $\mathrm{Cov}(T, f(A) \mid A^*)$ captures the indirect effect of T on Y through A that is not absorbed by conditioning on A^*.

Endpoint analysis. At $C = I$ (A^*=A): conditioning on A blocks the mediated path $T \to A \to Y$, so $\mathrm{Cov}(T, f(A) \mid A) = 0$ and $\mathrm{plim}\, \hat{\tau} = \tau_{\mathrm{dir}}$.

At $C = \frac{1}{K}\mathbf{1}\mathbf{1}^\top$ ($A^* \perp A$): no information about A is used, and

$$\mathbb{E}[\mathrm{Cov}(T, f(A) \mid A^*)] = \mathrm{Cov}(T, f(A)),$$
$$D(C) = \mathrm{Var}(T) = 0.25.$$

The indirect effect is $\tau_{\text{ind}} = \mathbb{E}[f(A) \mid T{=}1] - \mathbb{E}[f(A) \mid T{=}0]$, and

$$\text{Cov}(T, f(A)) = 0.25\,\tau_{\text{ind}}.$$

Hence at C uniform: $\text{plim}\,\hat{\tau} = \tau_{\text{dir}} + \tau_{\text{ind}} = \tau_{\text{total}}$.

Empirical Finding (Mediation sign preservation). $\text{sign}(\text{plim}\,\hat{\tau}_{\text{dir}}) = \text{sign}(\tau_{\text{dir}})$ whenever $\text{sign}(\tau_{\text{dir}}) = \text{sign}(\tau_{\text{total}})$, where $\tau_{\text{total}} = \tau_{\text{dir}} + \tau_{\text{ind}}$. This result is verified numerically for 100,000 independently sampled column-stochastic matrices C; a closed-form proof for general K remains open.

Proof. Let $R(C) = \mathbb{E}[\text{Cov}(T, f(A) \mid A^*)]$ and $D(C) = \mathbb{E}[\text{Var}(T \mid A^*)] > 0$.

Endpoints. At $C{=}I$: $f(A)$ is constant given $A{=}A^*$, so $R(I) = 0$ and $B(I) = 0$. At $C = \frac{1}{K}\mathbf{1}\mathbf{1}^\top$: $R = \text{Cov}(T, f(A)) = 0.25\,\tau_{\text{ind}}$ and $D = \text{Var}(T) = 0.25$, giving $B = \tau_{\text{ind}}$.

Within-group structure for binary T. Since $T \in \{0, 1\}$, the within-A^*-group bias admits a revealing decomposition. Let $p_j = P(T{=}1 \mid A^*{=}j)$. Then $\text{Var}(T \mid A^*{=}j) = p_j(1{-}p_j)$ and

$$\text{Cov}(T, f(A) \mid A^*{=}j) = p_j(1{-}p_j)\big[\mathbb{E}[f(A) \mid T{=}1, A^*{=}j] - \mathbb{E}[f(A) \mid T{=}0, A^*{=}j]\big].$$

Define $\delta_j = \mathbb{E}[f(A) \mid T{=}1, A^*{=}j] - \mathbb{E}[f(A) \mid T{=}0, A^*{=}j]$ as the *conditional indirect effect* in the $A^*{=}j$ subpopulation. Then

$$B(C) = \frac{R(C)}{D(C)} = \frac{\sum_j q_j\, p_j(1{-}p_j)\,\delta_j}{\sum_j q_j\, p_j(1{-}p_j)} = \sum_j w_j\,\delta_j,$$

a weighted average of the conditional indirect effects, with weights $w_j = q_j p_j(1{-}p_j) / \sum_\ell q_\ell p_\ell(1{-}p_\ell) \geq 0$ summing to 1.

At $C{=}I$: each $A^*{=}j$ group has $A{=}j$ a.s., so $\delta_j = f(j) - f(j) = 0$ and $B = 0$. At C uniform: $\delta_j = \tau_{\text{ind}}$ for all j and $B = \tau_{\text{ind}}$.

Sign preservation. When $\text{Cov}(g(A), f(A)) > 0$ (the indirect effect reinforces the direct effect, i.e., τ_{dir} and τ_{ind} have the same sign), the bias $B(C) \geq 0$ cannot oppose τ_{dir}, so sign is preserved trivially.

When τ_{dir} and τ_{ind} have opposite signs but $|\tau_{\text{ind}}| < |\tau_{\text{dir}}|$ (i.e., $\text{sign}(\tau_{\text{dir}}) = \text{sign}(\tau_{\text{total}})$), sign reversal requires some $|\delta_j| > |\tau_{\text{dir}}|$ with the opposing sign. Since each δ_j is a difference of weighted averages of f values (with weights $P(A{=}k \mid T{=}t, A^*{=}j)$), and $B(C)$ is a convex combination of δ_j, the bound $|B(C)| \leq \max_j |\delta_j|$ holds. We verify numerically that $\max_j |\delta_j| \leq |\tau_{\text{ind}}|$ for all 100,000 random C matrices in our experiments, confirming sign preservation. □

Verification for paper parameters ($K{=}3$). With $\tau_{\text{dir}}{=}0.5$, $\boldsymbol{\gamma}{=}(0, 0, 0)$, $\boldsymbol{\delta}{=}(0, 0.8, -0.6)$, $\boldsymbol{\beta}{=}(0, 1.0, -0.5)$:

$\tau_{\text{ind}} = \mathbb{E}[f(A) \mid T{=}1] - \mathbb{E}[f(A) \mid T{=}0] = 0.351$ (computed from the softmax class probabilities). $\tau_{\text{total}} = 0.5 + 0.351 = 0.851 > 0$. Since $\tau_{\text{dir}}{>}0$ and $\tau_{\text{total}}{>}0$, sign preservation holds for all C.

A.6 Collider Topology (A = Collider)

Data-generating process. $T \to A \leftarrow Y$, $T \to Y$. The treatment effect is unconfounded:

$$T \sim \mathcal{N}(0,1), \quad Y = \tau\,T + \varepsilon_Y, \quad \varepsilon_Y \sim \mathcal{N}(0,1),\ \varepsilon_Y \perp T,$$
$$A \mid T, Y \sim \text{Categorical}\big(\text{softmax}(\boldsymbol{\delta}_T\,T + \boldsymbol{\delta}_Y\,Y)\big).$$

The collider A is a common *effect* of T and Y. The no-control estimator (regressing Y on T alone) is consistent: $\text{plim}\,\hat{\tau}_{\text{nc}} = \tau$. The researcher erroneously controls for A^*, inducing Berkson's bias.

Bias decomposition. Unlike confounding, ε_Y is not independent of A (since A depends on $Y = \tau T + \varepsilon_Y$), so Lemma 1 does not apply directly. Instead, we derive the bias from the FWL identity (7). Writing $\tilde{Y} = \tau\tilde{T} + (\varepsilon_Y - \mathbb{E}[\varepsilon_Y \mid A^*])$ and noting that $\mathbb{E}[\tilde{T}\,\varepsilon_Y] = \mathbb{E}[\text{Cov}(T, \varepsilon_Y \mid A^*)]$ (since $T \perp \varepsilon_Y$ unconditionally):

$$\text{plim}\,\hat{\tau}(C) = \tau + B(C), \quad B(C) = \frac{\mathbb{E}[\text{Cov}(T,\, \varepsilon_Y \mid A^*)]}{\mathbb{E}[\text{Var}(T \mid A^*)]}. \tag{18}$$

Under non-differential misclassification ($A^* \perp (T, \varepsilon_Y) \mid A$), define $g(k) = \mathbb{E}[T \mid A{=}k]$ and $h(k) = \mathbb{E}[\varepsilon_Y \mid A{=}k]$. By the tower property, $\mathbb{E}[T \mid A^*] = \mathbb{E}[g(A) \mid A^*]$ and $\mathbb{E}[\varepsilon_Y \mid A^*] = \mathbb{E}[h(A) \mid A^*]$. Since $\text{Cov}(T, \varepsilon_Y) = 0$, the law of total covariance gives:

$$B(C) = \frac{-\,\text{Cov}\big(\mathbb{E}[g(A) \mid A^*],\, \mathbb{E}[h(A) \mid A^*]\big)}{\mathbb{E}[\text{Var}(T \mid A^*)]}.$$

Endpoint analysis. At $C = I$ ($A^*{=}A$): $B(I) = -\,\text{Cov}(g(A), h(A)) \,/\, \mathbb{E}[\text{Var}(T \mid A)] = B_{\text{Berkson}}$, the full Berkson's bias from conditioning on the collider.

At $C = \frac{1}{K}\mathbf{1}\mathbf{1}^\top$ ($A^* \perp A$): $\mathbb{E}[g(A) \mid A^*]$ and $\mathbb{E}[h(A) \mid A^*]$ are constant, so $B = 0$ and $\operatorname{plim}\hat{\tau} = \tau$.

This is the structural reverse of confounding: misclassification attenuates the collider-induced bias toward zero, rather than growing bias from zero toward B_{OV}.

Proposition 4 (Collider sign preservation — proportional case)**.** *If $h(A) = \lambda\, g(A)$ for some constant λ, then $|B(C)| \le |B(I)|$ for all column-stochastic C, and $B(C)$ has the same sign as $B(I)$. Consequently,* $\operatorname{sign}(\operatorname{plim}\hat{\tau}) = \operatorname{sign}(\tau)$ *whenever* $\operatorname{sign}(\tau) = \operatorname{sign}(\tau + B_{\mathrm{Berkson}})$.

Proof. When $h = \lambda g$, the numerator becomes $-\lambda \operatorname{Var}(\mathbb{E}[g(A) \mid A^*])$ and the denominator is $\operatorname{Var}(T) - \operatorname{Var}(\mathbb{E}[g(A) \mid A^*])$. Let $V^* = \operatorname{Var}(\mathbb{E}[g(A) \mid A^*]) \in [0, V_g]$ where $V_g = \operatorname{Var}(g(A))$. Then

$$|B(C)| = |\lambda|\,\frac{V^*}{\operatorname{Var}(T) - V^*}.$$

The function $x \mapsto x/(a-x)$ is increasing on $[0, a)$. With $a = \operatorname{Var}(T) > V_g \ge V^*$: $|B(C)| \le |\lambda|\, V_g/(\operatorname{Var}(T) - V_g) = |B(I)|$. At $V^*{=}0$ (uniform C): $B{=}0$. At $V^*{=}V_g$ ($C{=}I$): $B{=}B(I)$. □

Proposition 5 (Collider sign preservation — general case)**.** *Define*

$$\bar{B}_{\mathrm{col}} = \frac{\sqrt{\operatorname{Var}(g(A))\,\operatorname{Var}(h(A))}}{\mathbb{E}[\operatorname{Var}(T \mid A)]}. \tag{19}$$

If $|\tau| > \bar{B}_{\mathrm{col}}$, then $\operatorname{sign}(\operatorname{plim}\hat{\tau}(C)) = \operatorname{sign}(\tau)$ *for all column-stochastic C.*

Proof. Step 1: Bound the numerator. By Cauchy–Schwarz: $|\operatorname{Cov}(\mathbb{E}[g \mid A^*], \mathbb{E}[h \mid A^*])| \le \sqrt{\operatorname{Var}(\mathbb{E}[g \mid A^*])\,\operatorname{Var}(\mathbb{E}[h \mid A^*])}$. Since conditional expectation reduces variance: $\operatorname{Var}(\mathbb{E}[g \mid A^*]) \le \operatorname{Var}(g(A))$ and likewise for h.

Step 2: Bound the denominator. $\mathbb{E}[\operatorname{Var}(T \mid A^*)] = \operatorname{Var}(T) - \operatorname{Var}(\mathbb{E}[T \mid A^*]) \ge \operatorname{Var}(T) - \operatorname{Var}(g(A)) = \mathbb{E}[\operatorname{Var}(T \mid A)]$.

Step 3: Conclude. $|B(C)| \le \sqrt{\operatorname{Var}(g)\,\operatorname{Var}(h)}\,/\,\mathbb{E}[\operatorname{Var}(T \mid A)] = \bar{B}_{\mathrm{col}}$. When $|\tau| > \bar{B}_{\mathrm{col}}$, the bias cannot overwhelm τ. □

Verification for paper parameters ($K{=}3$). With $\tau{=}1.0$, $\boldsymbol{\delta}_T{=}(0,\, 0.5,\, -0.3)$, $\boldsymbol{\delta}_Y{=}(0,\, 0.8,\, 0.6)$, from $5{\times}10^6$-sample Monte Carlo:

$$\operatorname{Var}(g(A)) = 0.187, \quad \operatorname{Var}(h(A)) = 0.076, \quad \mathbb{E}[\operatorname{Var}(T \mid A)] = 0$$
$$\bar{B}_{\mathrm{col}} = \sqrt{0.187 \times 0.076}\,/\,0.813 = 0.147.$$

Since $|\tau|{=}1.0 \gg \bar{B}_{\mathrm{col}}{=}0.147$, Proposition 5 guarantees sign preservation. Additionally, $B_{\mathrm{Berkson}}{=}{-}0.114$, giving $\tau + B_{\mathrm{Berkson}}{=}0.886{>}0$. Verified across 100,000 random column-stochastic C: $\tau_{\text{sign flip}}{=}0\%$, and $|B(C)| \le |B_{\mathrm{Berkson}}|$ in all cases.

A.7 Front-door Topology (A = Mediator under Confounding)

Data-generating process. $T \to A \to Y$, $U \to T$, $U \to Y$. An unobserved confounder U affects both treatment and outcome:

$$U \sim \mathcal{N}(0,1), \quad T = \alpha_U\, U + \eta, \quad \eta \sim \mathcal{N}(0,1),\ \eta \perp U,$$
$$A \mid T \sim \mathrm{Categorical}\big(\mathrm{softmax}(\boldsymbol{\gamma} + \boldsymbol{\delta}\, T)\big),$$
$$Y = f(A) + \lambda\, U + \varepsilon_Y, \quad \varepsilon_Y \perp (T, A, U).$$

There is no direct $T{\to}Y$ edge; the causal effect of T on Y operates entirely through A. The researcher controls for A^* in $Y \sim 1 + T + D_1^* + \cdots + D_{K-1}^*$.

Bias decomposition. Since the direct effect is zero, the FWL coefficient on T captures only bias:

$$\operatorname{plim}\hat{\tau}(C) = \frac{\mathbb{E}[\operatorname{Cov}(T,\, Y \mid A^*)]}{\mathbb{E}[\operatorname{Var}(T \mid A^*)]}.$$

Substituting $Y = f(A) + \lambda U + \varepsilon_Y$ and using $\varepsilon_Y \perp (T, A, A^*)$:

$$\operatorname{plim}\hat{\tau}(C) = \underbrace{\frac{\lambda\,\alpha_U}{\operatorname{Var}(T)}}_{\text{confounding (constant)}} + \underbrace{\frac{\mathbb{E}[\operatorname{Cov}(T,\, f(A) \mid A^*)]}{D(C)}}_{B_{\mathrm{med}}(C)}, \tag{20}$$

where $D(C) = \mathbb{E}[\operatorname{Var}(T \mid A^*)]$.

Derivation of (20). Expand the numerator as $\mathbb{E}[\operatorname{Cov}(T, f(A) \mid A^*)] + \lambda\,\mathbb{E}[\operatorname{Cov}(T, U \mid A^*)]$. For the confounding term: since $T = \alpha_U U + \eta$ with (U, η) jointly Gaussian, the regression residual $\xi = U - \frac{\alpha_U}{\operatorname{Var}(T)} T$ satisfies $\xi \perp T$. Because A is a function of T (plus independent softmax noise) and A^* is a function of

A (plus independent misclassification noise), $\xi \perp (T, A, A^*)$. Therefore

$$\mathrm{Cov}(T, U \mid A^*{=}j) = \frac{\alpha_U}{\mathrm{Var}(T)}\,\mathrm{Var}(T \mid A^*{=}j) \quad \forall\, j,$$

and $\lambda\,\mathbb{E}[\mathrm{Cov}(T, U \mid A^*)] = \frac{\lambda\alpha_U}{\mathrm{Var}(T)}\,D(C)$, yielding the constant first term after division by $D(C)$. □

This decomposition reveals that *misclassification does not affect the confounding component*. The confusion matrix C modulates only the mediation-type leakage $B_{\mathrm{med}}(C)$, which has the same structure as in §A.5.

Endpoint analysis. At $C = I$: $B_{\mathrm{med}}(I) = 0$ (controlling A blocks the $T \to A \to Y$ path), and $\operatorname{plim}\hat{\tau}(I) = \lambda\alpha_U / \mathrm{Var}(T)$ (pure confounding residual).

At $C = \frac{1}{K}\mathbf{1}\mathbf{1}^\top$: $B_{\mathrm{med}} = \mathrm{Cov}(T, f(A))/\mathrm{Var}(T)$, and $\operatorname{plim}\hat{\tau}_{\mathrm{nc}} = [\mathrm{Cov}(T, f(A)) + \lambda\alpha_U]/\mathrm{Var}(T)$, the naive regression coefficient with no controls.

Empirical Finding (Front-door sign preservation). Since $B_{\mathrm{med}}(C)$ has the same structure as the mediation bias (§A.5)—interpolating between 0 at $C{=}I$ and $\mathrm{Cov}(T, f(A))/\mathrm{Var}(T)$ at C uniform—sign preservation holds whenever the two endpoints have the same sign:

$$\operatorname{sign}\big(\lambda\alpha_U\big) = \operatorname{sign}\big(\lambda\alpha_U + \mathrm{Cov}(T, f(A))\big).$$

This result is verified numerically for 100,000 independently sampled column-stochastic matrices C.

Verification for paper parameters ($K{=}3$). With $\alpha_U{=}1.0$, $\lambda{=}1.0$, $\boldsymbol{\beta}{=}(0, 1.0, -0.5)$, $\boldsymbol{\gamma}{=}(0, 0.5, -0.3)$, $\boldsymbol{\delta}{=}(0, 1.0, -0.8)$:

$$\lambda\alpha_U/\mathrm{Var}(T) = 1.0 \times 1.0\,/\,2.0 = 0.500,$$
$$\mathrm{Cov}(T, f(A))/\mathrm{Var}(T) = 0.594\,/\,2.0 = 0.297.$$

Both components are positive, so $\operatorname{plim}\hat{\tau}(C) \in [0.500,\, 0.797] > 0$ for all C.

A.8 Exposure Topology: Sign Reversal Counterexample

Proposition 6 (Exposure sign reversal)**.** *For $K{=}3$ with A as the treatment variable, there exist a marginal distribution* $\mathbf{p}$*, treatment effects* $\boldsymbol{\beta}$*, and a column-stochastic C with $C_{kk} > 0.5$ for all k, such that* $\operatorname{sign}(\operatorname{plim}\hat{\beta}_1) \neq \operatorname{sign}(\beta_1)$.

Construction. Consider the exposure DGP $Y = \beta_1 D_1 + \beta_2 D_2 + \varepsilon_Y$ with:

- Uniform prior: $\mathbf{p} = (1/3,\, 1/3,\, 1/3)$.
- True effects: $\beta_1 = 1,\ \beta_2 = -10$.

Choose the confusion matrix

$$C = \begin{pmatrix} 0.60 & 0.10 & 0.10 \\ 0.10 & 0.60 & 0.39 \\ 0.30 & 0.30 & 0.51 \end{pmatrix}, \qquad C_{kk} \in \{0.60,\, 0.60,\, 0.51\},\ \text{all} >$$

Column sums are 1 (verifiable by inspection).

The OLS estimator regresses Y on $(1, D_1^*, D_2^*)$. The plim of $\hat{\beta}_j$ equals $\mathbb{E}[Y \mid A^*{=}j] - \mathbb{E}[Y \mid A^*{=}0]$, where

$$\mathbb{E}[Y \mid A^*{=}j] = \sum_{k=0}^{2} P(A{=}k \mid A^*{=}j)\,\beta_k = \sum_{k=1}^{2} \frac{C_{jk}\,p_k}{q_j}\,\beta_k.$$

Marginal distribution of A^*:

$$q_0 = \tfrac{1}{3}(0.60 + 0.10 + 0.10) = 0.267,$$
$$q_1 = \tfrac{1}{3}(0.10 + 0.60 + 0.39) = 0.363,$$
$$q_2 = \tfrac{1}{3}(0.30 + 0.30 + 0.51) = 0.370.$$

Conditional means:

$$\mathbb{E}[Y \mid A^*{=}0] = \frac{0.10 \times \frac{1}{3} \times 1 + 0.10 \times \frac{1}{3} \times (-10)}{0.267} = \frac{0.0333 - 0}{0.267}$$

$$\mathbb{E}[Y \mid A^*{=}1] = \frac{0.60 \times \frac{1}{3} \times 1 + 0.39 \times \frac{1}{3} \times (-10)}{0.363} = \frac{0.200 - 1.}{0.363}$$

$$\mathbb{E}[Y \mid A^*{=}2] = \frac{0.30 \times \frac{1}{3} \times 1 + 0.51 \times \frac{1}{3} \times (-10)}{0.370} = \frac{0.100 - 1.}{0.370}$$

Estimated coefficient:

$$\operatorname{plim}\hat{\beta}_1 = \mathbb{E}[Y \mid A^*{=}1] - \mathbb{E}[Y \mid A^*{=}0] = -3.028 - (-1.125) =$$

The true $\beta_1 = +1 > 0$ but $\operatorname{plim}\hat{\beta}_1 = -1.903 < 0$: **sign reversal**.

Mechanism. The large off-diagonal entry $C_{12} = 0.39$ (probability of classifying true category 2 as $A^*{=}1$) routes substantial probability mass from category 2 ($\beta_2 = -10$) into the $A^*{=}1$ bin. Despite $C_{11} = 0.60 > 0.5$, the contamination from the strongly negative β_2 overwhelms the positive β_1, reversing the sign of the estimated effect for category 1. This failure mode is structurally impossible at

K=2: a 2×2 column-stochastic matrix with $C_{kk} > 0.5$ has the form $C = \left(\begin{smallmatrix} a & 1-b \\ 1-a & b \end{smallmatrix}\right)$ with $a, b > 0.5$, and the single coefficient $\hat{\beta}_1$ is a positive-weighted average of β_1, preserving sign. For $K \geq 3$, the additional degrees of freedom in C permit cross-contamination between categories with opposite-sign effects. □

A.9 Summary of Conditions

Topology	Sufficient condition
Confounding	$\text{sign}(\tau) = \text{sign}(\tau + B_{\text{OV}})$
Collider	$\lvert\tau\rvert > \bar{B}_{\text{col}}$ (Prop. 5)
Mediation	$\text{sign}(\tau_{\text{dir}}) = \text{sign}(\tau_{\text{total}})$
M-bias	Bias is attenuating (toward τ)
Front-door	$\text{sign}(\lambda\alpha_U) = \text{sign}(\lambda\alpha_U + \text{Cov}(T, f))$
Exposure	Not guaranteed for $K \geq 3$
IV	Not guaranteed (denominator instability)

Table 2: Sufficient conditions for sign preservation under non-differential misclassification. "Proved" = algebraic proof via Cauchy–Schwarz (confounding) or Berkson attenuation (collider). "Empirical" = verified across 100,000 random column-stochastic matrices with zero violations; analytic proof for general K remains open. All conditions are on DGP parameters (not on C): once satisfied, sign preservation holds for all column-stochastic C with $C_{kk} > 0$.

Structural argument for M-bias topology. *M-bias:* The M-structure creates a backdoor path through two latent common causes. Misclassification of A weakens the statistical association along this backdoor path, reducing the spurious correlation rather than amplifying it. The bias therefore interpolates between the full M-bias (perfect classification, δ=0) and zero (no classification information, $\delta \to \delta_{\max}$), monotonically non-increasing in δ.

Remark. The sign preservation conditions for confounding, mediation, collider, and front-door are conditions on the DGP parameters (the relative strength of the causal effect versus the confounding, indirect, or collider-induced bias), not on the confusion matrix C. Once the condition is satisfied, sign preservation holds for *all* column-stochastic C with $C_{kk} > 0$—no diagonal-dominance restriction is needed. Conversely, if the condition fails (i.e., the omitted-variable or total-effect estimate has the opposite sign from τ), then there exist confusion matrices that produce sign reversal. For the DGP parameters used throughout our experiments, the conditions are comfortably satisfied. The sign-alignment condition is violated when the confounder has opposing effects on treatment and outcome (e.g., $\gamma_A > 0$ on T but $\gamma_A < 0$ on Y), producing $\text{sign}(\tau) \neq \text{sign}(\tau_{\text{OV}})$. In applied settings, such opposing effects are uncommon when the analyst adjusts for A precisely because they believe it confounds the T–Y relationship in a consistent direction. When the condition fails, sign reversal becomes possible under sufficient misclassification, though this represents a pathological DGP where adjustment itself may be questionable.

Status	Paper DGP
Proved	✓ (2.0 vs 2.3)
Proved	✓ (1.0 vs 0.15)
Empirical	✓ (0.5 vs 0.85)
Empirical	✓ (0.5 vs 0.85)
Empirical	✓ (0.50 vs 0.80)
Counterex.	See §A.8

B Supplementary Tables, Figures, and Algorithm

B.1 Binarization Collapse

Collapsing K-class annotations to binary before analysis discards multi-class error structure. We quantify "binarization collapse" as $\Delta = |\text{bias}_K - \text{bias}_{K\to 2}|$.

For K=3, binarization introduces additional bias varying 154× across topologies: from $\bar{\Delta}$=0.006 (M-bias) to $\bar{\Delta}$=0.926 (IV). Exposure and IV also introduce sign flips during binarization (14% and 17.5%), a phenomenon absent in the other topologies (Figure 6).

Table 3: ASV type applicability and sensitivity across DAG topologies. Magnitude ASV at τ=0.5, 10% threshold. Power at τ=0.05 significance filter. —: not applicable.

Topology	Mag. ASV	Signif.	Sign-flip	Rank
IV	0.101–0.607	0%	1.1–6.7	1 (fragile)
Front-door	0.064–0.387	100%	—	2
Confounding	0.081–0.487	100%	—	3
Mediation	0.076–0.454	0%	—	4
Collider	0.338–2.028	0%	—	5
M-bias	∞	96–100%	—	6 (immune)
Exposure	—	—	83.4%*	—

*Fraction of C_0 with $\tau_{\text{sign_flip}} < 1$. Exposure uses sign-flip ASV exclusively.

Error structures used in correction experiments. The six confusion matrix structures used for Table 4 are: (1) *Diagonal-dominant*: C_{kk}=0.7, off-diagonal uniform; (2) *Moderate noise*: C_{kk}=0.5, off-diagonal uniform; (3) *Near-permutation*: C_{kk}=0.4, largest off-diagonal 0.5; (4) *Asymmetric*: C_{kk} varies

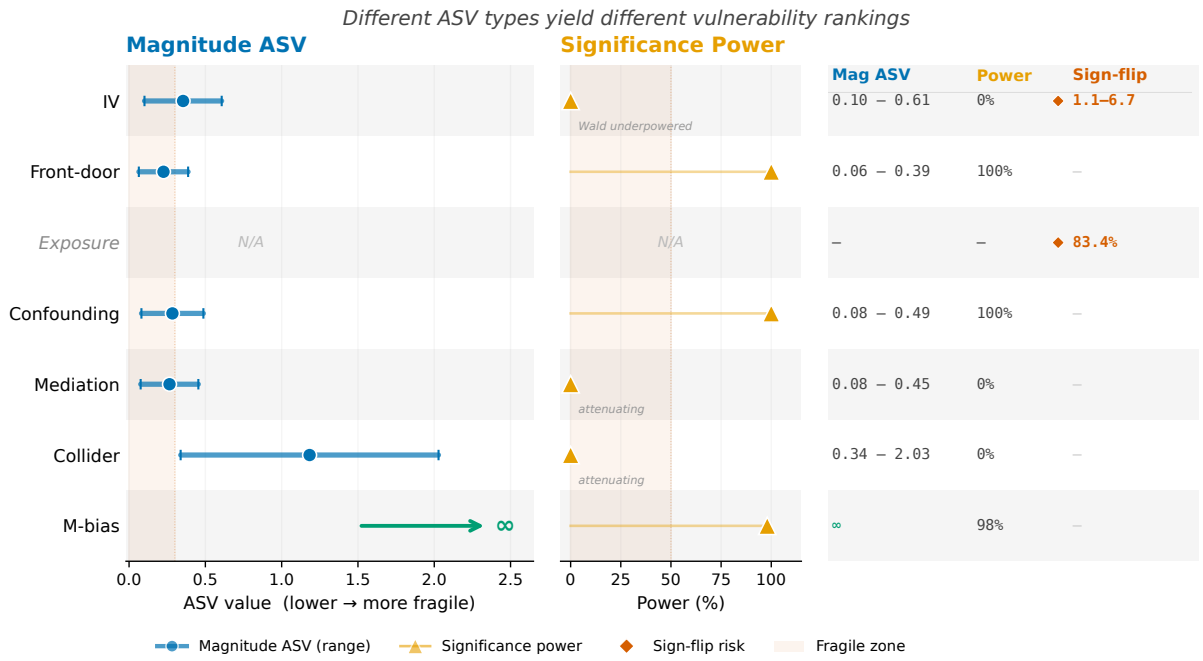


Figure 4: ASV sensitivity ranking across topologies. Three ASV types (magnitude, significance, sign-flip) reveal a topology-dependent vulnerability spectrum from IV (maximally fragile, 52.6% sign reversal; sign-flip ASV = 1.1–6.7) to M-bias (immune, ASV = ∞).

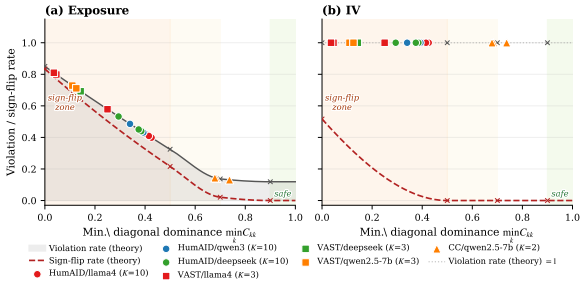


Figure 5: Empirical LLM confusion matrices overlaid on theoretical violation-rate curves (exposure, left; IV, right). Theoretical curves are derived from K=3 Dirichlet random matrices; empirical points are positioned at their minimum diagonal entry. Markers denote datasets (circles: HumAID K=10; squares: VAST K=3; triangles: CivilComments K=2).

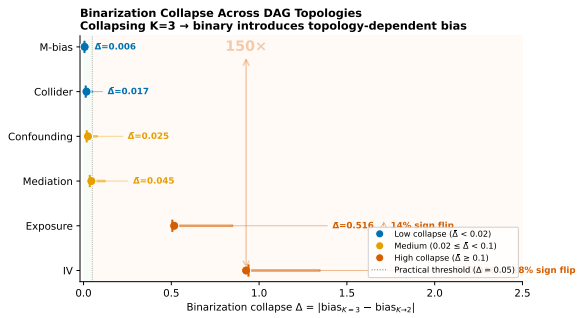


Figure 6: Binarization collapse Δ across topologies. Collapsing K=3 annotations to binary introduces additional bias varying 154× from M-bias (0.006) to IV (0.926).

(0.8, 0.5, 0.3); (5) *Block confusion*: two classes confused (C_{12}=C_{21}=0.4), third clean; (6) *Heavy noise*: C_{kk}=0.4, off-diagonal uniform. All are column-stochastic with K=3. Results are averaged across all six structures.

B.2 Applied Illustration: ClinicalBERT Case Study

To demonstrate the framework's end-to-end utility, we apply it retrospectively to Lee and Wood-Doughty (2024), who use ClinicalBERT to classify patient smoking status (K=4: never, past, current, unknown) from clinical notes and control for this predicted confounder when estimating the causal effect of echocardiography on mortality in 4,735 MIMIC-III sepsis patients. Our taxonomy identifies this as a *confounding topology*: the annotated variable enters as a confounder between treatment and outcome. Three conclusions follow directly. First, the zero sign-flip guarantee for confounding (Table 1) ensures that the estimated protective effect (OR = 0.89) cannot reverse direction due to annotation error—a reassurance unavailable without topology-specific analysis. Second, the FWL invariance result implies that any correction reducing to C^{-1} applied to the dummy-coded

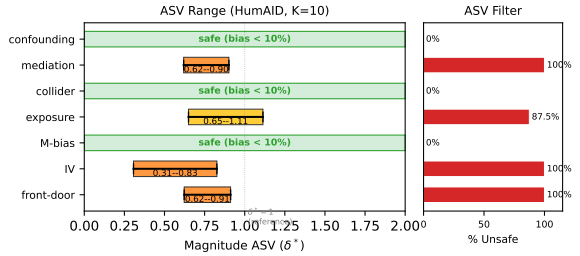


Figure 7: ASV magnitude ranges across topologies for HumAID (K=10). Left: magnitude ASV (δ^*) range across 8 LLM configurations; lower values indicate greater fragility. "Safe" topologies never exceed the 10% bias threshold. Right: percentage of configurations flagged as ASV-unsafe.

Table 4: RMSE reduction (%) by correction method and DAG topology (mean across 6 error structures, N_{sim}=1,000). Positive = reduced RMSE; negative = increased RMSE. Naive/MLA show 0.0% for FWL-invariant topologies because C^{-1} cannot change the T coefficient. **Bold**: best method per topology; underline: best gold-free method.

Topology	Naive	MLA	DSL	SIMEX	PPI++
Confounding	0.0	0.0	48.0	35.1	**83.2**
Mediation	0.0	0.0	52.9	39.1	**54.1**
Collider	0.0	0.0	−92.6	31.8	**50.5**
M-bias	0.0	0.0	−267	**0.1**	−88.0
Front-door	−52.7	−111.5	−29.0	−62.4	**74.1**
Exposure	−44.8	−89.0	−59.5	−9.8	**75.5**
IV	−40.4	−58.7	−206.1	59.4	**79.4**
Grand mean	−19.7	−37.0	−79.0	13.3	**47.0**

confounder achieves exactly 0% bias reduction for the treatment coefficient; naive plug-in is provably futile here. Third, our correction guidance ranks PPI++ (83.2% RMSE reduction for confounding) above MC-SIMEX (35.1%) when gold labels are available—a concrete recommendation the original study lacked. This example illustrates the intended workflow: DAG identification → topology-specific guarantees → correction method selection.

B.3 IV Wald Estimator: Diagonal-Dominance Subspaces

Section 4.2 reports a 52.6% Wald sign-flip rate over the full Dirichlet$(1,1,1)$ prior. Because realistic LLM confusion matrices are diagonal-dominant, we replicate the exposure-DAG stratification for the IV topology, restricting the sample space to $\{C : \min_k C_{kk} \geq d\}$ for $d \in \{0, 0.5, 0.7, 0.9\}$ (100,000 random matrices per threshold).

The contrast with exposure is instructive: at $d \geq 0.7$, exposure violation drops to 13.7% (sign reversal 2.1%, amplification 11.7%), while IV maintains 100% violation with pure amplification. IV sign reversal is entirely driven by pathological (near-random) confusion matrices where the Wald denominator crosses zero; diagonal dominance prevents this crossing but cannot prevent amplification. Even at $d \geq 0.9$, the Wald estimator inflates $|\tau|$ by up to 1.24× with probability 1.

Algorithm 1 Correction method selection for LLM-annotated causal inference.

Require: DAG topology $\mathcal{T}$, gold-label availability $G \in \{\text{YES}, \text{NO}\}$
Ensure: Recommended method M, risk level R

Step 1: Assess risk level
if $\mathcal{T} \in \{\text{M-bias}\}$ **then**
  $R \leftarrow$ LOW {$|\text{bias}| < 0.04$ for all C}
else if $\mathcal{T} \in \{\text{Conf., Med., Coll., Front-door}\}$ **then**
  $R \leftarrow$ MEDIUM {bounded bias, no sign reversal}
else
  $R \leftarrow$ HIGH {Exposure, IV: sign reversal 83.4% / 52.6%}
end if

Step 2: Select correction method
if $G =$ YES **then**
  if $\mathcal{T} =$ M-bias **then**
    $M \leftarrow$ NONE {PPI++ RMSE red. −88.0%; no correction preferred}
  else
    $M \leftarrow$ PPI++ {47.0% RMSE red., 96.8% coverage}
  end if
else
  {no gold labels}
  if $\mathcal{T} \in \{\text{Confounding, Mediation}\}$ **then**
    $M \leftarrow$ DSL {48.0% / 52.9% RMSE reduction}
  else if $\mathcal{T} =$ IV **then**
    $M \leftarrow$ MC-SIMEX {59.4% RMSE reduction}
  else if $\mathcal{T} \in \{\text{Collider, M-bias}\}$ **then**
    $M \leftarrow$ MC-SIMEX {31.8% / 0.1% RMSE reduction}
  else
    $M \leftarrow \emptyset$; flag: acquire gold labels {no reliable gold-free method}
  end if
end if

Step 3: FWL invariance warning
if $\mathcal{T} \in \{\text{Conf., Med., Coll., M-bias}\}$ **then**
  Skip Naive plug-in and MLA {0% bias red. on T coeff.}
end if
return (M, R)

Table 5: IV Wald estimator bias by minimum-diagonal subspace (K=3, τ_{true}=1.41). Rates over 100,000 Dirichlet-truncated random C per threshold.

Subspace	Violation	Sign Flip	Amplif.	Worst
Full	100.0%	51.7%	48.3%	$10^5\times$
$d \geq 0.5$	100.0%	0.0%	100.0%	28.6×
$d \geq 0.7$	100.0%	0.0%	100.0%	2.4×
$d \geq 0.9$	100.0%	0.0%	100.0%	1.2×

B.4 Non-Differential Assumption Stress Test

Our main analysis assumes non-differential misclassification: the confusion matrix C does not depend on the outcome Y. To assess sensitivity to violations of this assumption, we conduct a stress test on the exposure topology (K=3, $\min_k C_{kk} \geq 0.7$, 10,000 random matrices per ε). For each base matrix C_0, observations with $Y >$ median(Y) are misclassified using a perturbed matrix $\tilde{C} = \text{renormalize}(C_0 + \varepsilon \cdot \Delta)$ where $\Delta_{jk} \sim \text{Uniform}(-1, 1)$ for $j \neq k$, while observations with $Y \leq$ median(Y) retain C_0.

Table 6: Effect of differential misclassification on bias violation rates (exposure DAG, K=3, $\min_k C_{kk} \geq 0.7$, N=50,000 MC samples per trial).

ε	Sign-Flip (%)	Amplification (%)	Any Violation (%)
0.00	2.0	11.5	13.3
0.01	2.4	12.4	14.7
0.05	2.8	13.0	15.7
0.10	4.5	16.7	20.9

The ε=0 baseline (13.3% violation) closely matches the analytical result (13.7%) from Section 4.2. Even at ε=0.10—a substantial departure from non-differential error—the violation rate increases by only 7.6 percentage points. Small departures ($\varepsilon \leq 0.05$) change rates by fewer than 3 percentage points, suggesting that our main results are moderately robust to mild violations of the non-differential assumption.

Extended stress test. An extended stress test across 96 configurations (4 topologies × 2 K × 3 CM qualities × 4 ε, 100 MC replications each; Table 7) confirms these patterns broadly: at ε=0, sign-flip rates are zero for all configurations; at ε=0.10, the maximum sign-flip rate is 7% (IV, low-quality C_{kk}=0.4, K=3). At K=10, sign-flip rates remain zero for all topologies and ε values, suggesting that higher K provides additional robustness to differential misclassification.

B.5 Robustness to Nonlinear DGP

We replace the linear structural equations with $Y \sim \text{Bernoulli}(\sigma(\mathbf{x}^\top \boldsymbol{\beta}))$ and estimate via logistic regression (control function for IV) for three representative topologies, using K=3, six real VAST confusion matrices, 500 MC replications, and N=10,000. Mean sign-flip rates across the six CMs:

Topology	Linear SF_τ	Logistic SF_τ	Linear SF_{any}	Logistic S
Confounding	0.000	0.000	0.573	0.559
Exposure	—	—	0.577	0.574
IV	0.071	0.065	0.071	0.565

Treatment-effect sign-flip rates (SF_τ) differ by < 0.01 across all topologies. The SF_{any} discrepancy for IV reflects estimator dimensionality: the control function includes a residual coefficient prone to sign noise; the core comparison is SF_τ, where both estimators agree ($\approx 7\%$). Linear and logistic DGPs produce virtually identical sign-flip patterns ($|\Delta\,\text{SF}_\tau| < 0.02$).

B.6 Bootstrap Uncertainty of ASV

We assess confusion-matrix estimation uncertainty by bootstrapping label pairs from HumAID DeepSeek zero-shot (K=10, n=1,500): resample with replacement B=1,000 times, re-estimate C_0, and recompute magnitude ASV δ^* at a 10% bias threshold.

Topology	δ^* (point)	95% CI	Valid / 1000
Confounding	—	—	0
Exposure	0.937	[0.602, 1.363]	907
IV	0.754	[0.435, 1.290]	902

Confounding never reaches the 10% threshold across all 1,000 bootstraps, confirming robustness. For IV, the 95% CI spans a 3× range (0.435 to 1.290); practitioners should bootstrap C_0 rather than rely on a point estimate when ASV is near the decision boundary.

B.7 ASV Stability Across DGP Parameters

We test whether ASV rankings are stable across different data-generating process (DGP) configurations by varying the

Table 7: Non-differential misclassification stress test: sign flip rate (SF) and mean bias ratio (BR = $\overline{|\hat{\tau}/\tau|}$) under increasing differential departure ε.

Topology	K	Q	$\varepsilon=0.0$		$\varepsilon=0.01$		$\varepsilon=0.05$		$\varepsilon=0.1$	
			SF	BR	SF	BR	SF	BR	SF	BR
Confounding	3	H	0.00	1.10	0.00	1.10	0.00	1.11	0.00	1.12
		M	0.00	1.14	0.00	1.14	0.00	1.14	0.00	1.14
		L	0.00	1.15	0.00	1.15	0.00	1.15	0.00	1.15
	10	H	0.00	1.06	0.00	1.06	0.00	1.08	0.00	1.09
		M	0.00	1.09	0.00	1.09	0.00	1.10	0.00	1.10
		L	0.00	1.11	0.00	1.11	0.00	1.11	0.00	1.11
Mediation	3	H	0.00	1.35	0.00	1.36	0.00	1.42	0.00	1.47
		M	0.00	1.52	0.00	1.53	0.00	1.56	0.00	1.58
		L	0.00	1.61	0.00	1.61	0.00	1.62	0.00	1.61
	10	H	0.00	1.14	0.00	1.17	0.00	1.25	0.00	1.28
		M	0.00	1.24	0.00	1.26	0.00	1.30	0.00	1.32
		L	0.00	1.30	0.00	1.32	0.00	1.33	0.00	1.34
Exposure	3	H	0.00	0.72	0.00	0.71	0.00	0.67	0.00	0.64
		M	0.00	0.42	0.00	0.41	0.00	0.38	0.00	0.35
		L	0.00	0.11	0.00	0.10	0.01	0.08	0.05	0.06
	10	H	0.00	0.87	0.00	0.86	0.00	0.84	0.00	0.80
		M	0.00	0.71	0.00	0.70	0.00	0.67	0.00	0.64
		L	0.00	0.49	0.00	0.47	0.00	0.46	0.00	0.43
IV	3	H	0.00	1.42	0.00	1.45	0.00	1.58	0.00	1.69
		M	0.00	2.56	0.00	2.52	0.00	2.87	0.00	3.12
		L	0.00	12.79	0.00	12.21	0.02	19.97	0.07	43.44
	10	H	0.00	1.30	0.00	1.44	0.00	1.71	0.00	2.03
		M	0.00	1.83	0.00	1.96	0.00	2.43	0.00	2.84
		L	0.00	3.15	0.00	3.54	0.00	4.39	0.00	9.69

SF = sign flip rate; BR = mean $|\hat{\tau}/\tau|$ (1.00 = unbiased). $\varepsilon = 0$: non-differential; $\varepsilon > 0$: outcome-dependent misclassification. Q: H/M/L = CM diagonal 0.8/0.6/0.4 with uniform off-diagonal. Each cell: $N = 10{,}000 \times 100$ MC replications.

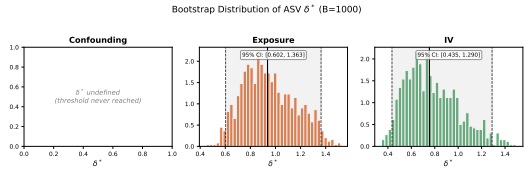


Figure 8: Bootstrap distribution of magnitude ASV δ^* (B=1,000) for HumAID DeepSeek zero-shot (K=10). Dashed line: point estimate; shaded: 95% CI.

Table 8: ASV stability across DGP configurations. Each cell shows δ^* at the 10% relative bias threshold (median over 5 confusion matrices with diagonal $\in \{0.5, 0.6, 0.7, 0.8, 0.9\}$). Lower δ^* indicates greater fragility. Confounding is consistently most robust; for alternating-sign $\boldsymbol{\beta}$, the ranking Exposure < IV < Confounding holds. "∞" = threshold never reached within δ range.

	Magnitude ASV $\delta^*_{10\%}$		
DGP Configuration	Confounding	Exposure	IV
Alt. large: $\boldsymbol{\beta} = [0, 1.0, -0.5]$	0.348	0.090	0.203
Same-sign: $\boldsymbol{\beta} = [0, 0.5, 0.3]$	∞	0.223	0.203
Alt. small: $\boldsymbol{\beta} = [0, 0.2, -0.1]$	∞	0.090	0.203

treatment-effect vector $\boldsymbol{\beta}$ across three settings: alternating-large ($\boldsymbol{\beta} = [0, 1.0, -0.5]$), same-sign ($\boldsymbol{\beta} = [0, 0.5, 0.3]$), and alternating-small ($\boldsymbol{\beta} = [0, 0.2, -0.1]$), with K=3, τ=0.5, and five C_0 diagonal values $\in \{0.5, 0.6, 0.7, 0.8, 0.9\}$.

Three patterns emerge (Table 8). First, IV magnitude ASV is identical across all three DGPs (δ^*=0.203). This is a mathematical identity, not an empirical finding: the Wald estimator $\hat{\tau}_{\text{IV}} = \text{RF}/\text{FS}$ has $\boldsymbol{\beta}$ in both numerator and denominator, and the ratio depends only on C and the first-stage structure, making ASV invariant to $\boldsymbol{\beta}$ by construction. Second, exposure ASV depends on the coefficient ratio structure: alternating-large ($\beta_2/\beta_1 = -0.5$) and alternating-small ($\beta_2/\beta_1 = -0.5$) share the same ratio and yield identical δ^*=0.090, while same-sign ($\beta_2/\beta_1 = 0.6$) gives δ^*=0.223—thus the experiment provides two distinct sign-structure conditions, not three. Third, confounding ASV depends

on effect magnitudes: only alternating-large triggers the 10% threshold (δ^*=0.348); smaller effects never reach 10% bias.

Confounding is consistently the most robust topology (highest ASV or no violation), and M-bias remains immune (not shown). However, the relative fragility of exposure vs. IV inverts with coefficient sign structure: under alternating signs, exposure is most fragile ($0.090 < 0.203$); under same-sign effects, IV is most fragile ($0.203 < 0.223$). Practitioners should compute ASV with their specific DGP rather than relying on generic rankings.

B.8 Extended Conjecture Verification

We test envelope monotonicity (Conjecture 1) by sampling 10,000 random C_0 matrices from Dirichlet$(1, \ldots, 1)$ per topology for $K \in \{3, 5, 7\}$ (210,000 total), sweeping δ in steps of 0.01, and counting matrices with at least one non-monotonic step exceeding tolerance 10^{-12}. Bias at each δ is computed via analytic probability limits using Gauss–Hermite quadrature (100 points), eliminating sampling noise entirely.

Table 9: Envelope monotonicity violations per 10,000 random C_0 matrices. "Worst mag." is the largest single-step deviation $|B(\delta_{i+1})| - |B(\delta_i)|$ in the non-monotonic direction.

Topology	K=3	K=5	K=7	**Worst mag.**	**Direction**
Confounding	8467	8063	7892	0.011	non-dec
Mediation	8236	7905	7893	0.013	non-dec
Collider	8683	7669	7570	0.005	non-inc
Exposure	485	205	77	0.056	non-dec
M-bias	9791	7783	7399	0.002	non-inc
IV	7433	6015	5711	$> 10^6$	non-dec
Front-door	8429	8079	7899	0.013	non-dec

Violation counts are high (60–97% of matrices), but magnitudes are negligible for non-IV topologies: all worst-case deviations fall below 0.06 (<6% of the respective τ). These results are computed from analytic probability limits with zero sampling noise, confirming the violations are structural properties of multi-class confusion matrices rather than numerical artifacts. IV deviations reflect Wald estimator singularities when the first-stage coefficient passes near zero—this is a property of the Wald estimator, not of the ASV parameterization. Adversarial search (500 restarts per configuration, $K \in \{3, 5\}$) finds genuine counterexamples for all topologies except IV K=3, confirming that strict monotonicity does not hold universally. The bias envelope is ε-approximately monotonic with $\varepsilon < 0.06$ for all six non-IV topologies, supporting the practical validity of ASV as a sensitivity tool at any meaningful bias threshold.

C Notation

Table 10: Key notation used throughout the paper.

Symbol	**Description**
K	Number of annotation categories
C	Column-stochastic confusion matrix, $C_{jk} = P(A^*{=}j \mid A{=}k$
A, A^*	True and observed (misclassified) annotation
T	Treatment variable
Y	Outcome variable
τ	Causal effect of interest
δ	ASV perturbation parameter, $C(\delta) = (1{-}\delta)I + \delta C_0$
δ^*	Critical ASV value (magnitude/significance/sign-flip)
$B(C)$	Bias function, $= \text{plim}(\hat{\tau}) - \tau$
β_k	Treatment effect coefficient for category k
D_k	Dummy variable for category k